Supplementary Material: The Non-Stationary Landauer Efficiency

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1 S1. Full proof of Remark 6 (softmax-regularity for OU)

Remark 6 (§ 4.4) states: in the OU class of § 5.2 the exponential decorrelation of the autocovariance of the logits $\theta(t)$ is transferred through the softmax to any bounded measurable functional $\xi(\theta)$ (in particular, to the entries of the transition matrix $P_{ij}(\theta)$ and to any sufficient statistic of prediction). The proof has five steps: (i) Lipschitz softmax in ℓ_∞; (ii) transfer to the TV of the transition matrix; (iii) boundedness of the OU trajectory on the ball $R(\delta)$ with probability $1-\delta$; (iv) transfer of the decay to the covariance of ξ; (v) translation of the TV bound into a bound on the conditional MI.

1.1 S1.1 Step 1. Lipschitz continuity of softmax

Let softmax : $\mathbb{R}^k \to \Delta^{k-1}$ be the standard map $\mathrm{softmax}_i(\theta) = \exp\theta_i / \sum_j \exp\theta_j$. The Jacobian has the form $\partial\mathrm{softmax}_i/\partial\theta_j = \mathrm{softmax}_i(\delta_{ij} - \mathrm{softmax}_j)$; the diagonal entries $p_i(1-p_i) \geq 0$, the off-diagonal entries $-p_ip_j \leq 0$, while the sum of absolute values over a column does not exceed $2p_j$. The standard consequence is the ℓ_1/ℓ_∞ pairwise bound:

$$\|\mathrm{softmax}(\theta) - \mathrm{softmax}(\theta')\|_1 \;\leq\; 2\,\|\theta - \theta'\|_\infty. \tag{S1.1}$$

The derivation of (S1.1) is the integration of the Jacobian along the line $\theta(s) = \theta + s(\theta' - \theta)$, $s \in [0, 1]$, applying Hölder's inequality to each component of the gradient; the constant 2 is optimal. A standard result, with no additional assumptions on the dimension k.

1.2 S1.2 Step 2. Transfer to the total variation of the transition matrix

In the OU parametrisation of § 5.2 the rows of the transition matrix are given by a row-wise softmax over the logits: $P_{ij}(\theta) = \exp\theta_{ij}/\sum_l \exp\theta_{il}$, where by convention $\theta_{ii} \equiv 0$. For each row i, applying (S1.1) to the vectors $(\theta_{ij})_{j=1}^k$ and $(\theta'_{ij})_{j=1}^k$:

$$\|P_{i,\cdot}(\theta) - P_{i,\cdot}(\theta')\|_1 \;\le\; 2\max_j |\theta_{ij} - \theta'_{ij}|.$$

In terms of the TV distance between the distributions $P_{i,\cdot}(\theta)$ and $P_{i,\cdot}(\theta')$ (half the ℓ_1 distance, $\|\mu-\nu\|_{TV} = \frac{1}{2}\|\mu-\nu\|_1$) one obtains

$$\|P_{i,\cdot}(\theta) - P_{i,\cdot}(\theta')\|_{TV} \;\le\; \max_j |\theta_{ij} - \theta'_{ij}| \;\le\; \|\theta-\theta'\|_\infty. \tag{S1.2}$$

This gives a row-wise Lipschitz constant $L = 1$ for the softmax parametrisation of the transition matrix. The coordinate-wise bound (instead of the full $\|\theta-\theta'\|_\infty$ over all rows) is essential for step S1.4: the $K = k(k-1)$ logit components of the OU system are independent (the diagonal anchor $\theta_{ii} \equiv 0$), and each row of the matrix depends only on its own subsystem of $k-1$ components.

1.3 S1.3 Step 3. Boundedness of the OU trajectory with high probability

The OU process on each coordinate θ_{ij} in the stationary regime is Gaussian, $\mathcal{N}(0, \sigma^2/(2\lambda))$; the autocovariance $\mathbb{E}[\theta_{ij}(t)\theta_{ij}(s)] = (\sigma^2/(2\lambda))\, e^{-|t-s|/\tau_E}$ with $\tau_E = 1/\lambda$. By the concentration inequality for Lipschitz functionals of a Gaussian measure (Borell–Tsirelson, the standard result of the theory of Gaussian processes), for a single coordinate $\Pr[\sup_{[0,T]} |\theta_{ij}(t)| > R] \le 2\exp(-R^2\lambda/\sigma^2)$; the union bound over K coordinates gives the radius

$$R(\delta) \;=\; \sigma\sqrt{\ln(2K/\delta)/\lambda}, \tag{S1.3}$$

ensuring $\Pr[\sup_{[0,T],ij} |\theta_{ij}(t)| > R(\delta)] \le \delta$. On the event $\{\|\theta(t)\|_\infty \le R(\delta)\}$ the local Lipschitz constant of softmax (S1.2) applies, and, through the product of measures, for a pair of times (s,t) simultaneously.

1.4 S1.4 Step 4. Transfer of the exponential decay to $\boldsymbol{\xi}$

Let $\xi : \mathbb{R}^K \to \mathbb{R}$ be a bounded measurable function, $|\xi| \le B$, Lipschitz coordinate-wise with constant L_ξ in the ℓ_∞ metric: $|\xi(\theta) - \xi(\theta')| \le L_\xi \|\theta-\theta'\|_\infty$ (for the row-wise softmax projection $L_\xi \le 1$ by (S1.2)). The covariance of ξ in the stationary OU regime is estimated through its representation as a difference of products:

$$\mathrm{Cov}\big(\xi(\theta(t)),\, \xi(\theta(s))\big) \;=\; \mathbb{E}\big[(\xi(\theta(t)) - \bar\xi)(\xi(\theta(s)) - \bar\xi)\big],$$

where $\bar\xi = \mathbb{E}[\xi(\theta)]$ under the stationary measure. On the event of step S1.3 (of probability $\ge 1-\delta$) one has $|\xi(\theta(t)) - \bar\xi| \le L_\xi \cdot 2R(\delta)$ when compared with the deterministic value at $\theta = 0$ (the symmetry point of the stationary measure) plus a bounded contribution from the shift $\bar\xi$, absorbed into $|\xi| \le B$. Applying the Cauchy–Schwarz inequality and the standard estimate of the OU autocovariance $\mathbb{E}[\theta_{ij}(t)\theta_{ij}(s)] = (\sigma^2/(2\lambda))e^{-|t-s|/\tau_E}$ gives

$$\big|\mathrm{Cov}(\xi(\theta(t)), \xi(\theta(s)))\big| \;\le\; 2B \cdot L_\xi \cdot \frac{\sigma^2}{2\lambda} \cdot e^{-|t-s|/\tau_E} + 2B^2\delta, \tag{S1.4}$$

where the second term is the correction from the low-probability event of step S1.3, in which the trajectory exits the ball of radius $R(\delta)$. The choice $\delta = e^{-|t-s|/\tau_E}$ for the separation of events by the characteristic decorrelation time makes the δ-correction a term of the same order as the leading exponential term. Finally: for any bounded measurable ξ with coordinate-wise Lipschitz constant L_ξ,

$$\big|\mathrm{Cov}(\xi(\theta(t)), \xi(\theta(s)))\big| \;\le\; C_1(B, L_\xi, \sigma, \lambda) \cdot e^{-|t-s|/\tau_E}, \tag{S1.4'}$$

with explicit constant $C_1 = (BL_\xi\sigma^2/\lambda) + 2B^2$ (or smaller upon optimising the choice of δ). The exponential decay of the autocovariance of θ is transferred to ξ with the same characteristic time τ_E — this is precisely the substantive statement of Remark 6.

1.5 S1.5 Step 5. Transfer to the conditional mutual information

The chain "covariance of $\xi \to$ MI" via inverse Pinsker (Sason–Verdú-type) requires boundedness of the log-density ratio $\alpha = \log\|dP/dQ\|_\infty$, which for the softmax parametrisation does not hold universally: even on the ball $R(\delta)$ of step S1.3 it gives $\alpha \le 2R(\delta) + \log k$, not simply $\log k$. Therefore step S1.5 is built through *MI-tensorisation* over the independent coordinates of OU followed by DPI on ξ — a formally correct replacement. Direct χ^2-tensorisation is inapplicable here: χ^2 is *multiplicative* over the independence of coordinates ($1 + \chi^2_{\text{joint}} = \prod_{ij}(1 + \chi^2_{\text{coord}})$), not additive, so the naive inequality "$\chi^2_{\text{joint}} \le K\chi^2_{\text{coord}}$" is false for χ^2 and is circumvented through MI-tensorisation, which is additive precisely because MI is additive over independent coordinates.

Step 5a. KL $\le \chi^2$ as an auxiliary coordinate-wise estimate. The standard inequality (Cover and Thomas 2006, exercise 2.45): for any P_{XY}, P_X, P_Y

$$I(X;Y) \;=\; D_{KL}\big(P_{XY} \,\|\, P_X \otimes P_Y\big) \;\le\; \chi^2\big(P_{XY} \,\|\, P_X \otimes P_Y\big).$$

This inequality does not require boundedness of the density ratio; it holds in any σ-algebra. In the steps below it is applied *only coordinate-wise* to justify the asymptotics $I_{\text{coord}} = \frac{1}{2}\rho^2(1 + o(1))$, not for tensorisation.

Step 5b. MI for a bivariate Gaussian pair on a coordinate. For a single coordinate θ_{ij} in the stationary OU, the pair $(\theta_{ij}(t), \theta_{ij}(s))$ is Gaussian with normalised correlation $\rho(t, s) = e^{-|t-s|/\tau_E}$. The standard formula for the MI of a bivariate Gaussian distribution with marginal variance $\sigma_0^2 = \sigma^2/(2\lambda)$:

$$I\big(\theta_{ij}(t);\, \theta_{ij}(s)\big) \;=\; -\frac{1}{2}\ln\big(1 - \rho^2(t,s)\big). \tag{S1.5a}$$

For $|t - s| \ge \tau_E$ one has $\rho^2 \le e^{-2} < 0.14$, and the Taylor expansion gives $I_{\text{coord}} = \frac{1}{2}\rho^2 + O(\rho^4) \;\le\; \frac{1}{2}\rho^2(1 + o(1)) \;\le\; \frac{1}{2}e^{-2|t-s|/\tau_E}$. The alternative expression through χ^2 ($\chi^2_{\text{coord}} = \rho^2/(1 - \rho^2)$ — a direct computation) gives the same leading term $\frac{1}{2}\rho^2$

through step 5a with exponent $2/\tau_E$ (not $1/\tau_E$); MI and χ^2 on a coordinate agree asymptotically.

Step 5c. MI-tensorisation over K independent coordinates. MI is additive over statistically independent pairs (Cover and Thomas 2006, ch. 2.5): the OU coordinates are independent by construction (step S2.1: W_{ij} are independent; the covariance is diagonal), and therefore

$$I\big(\theta(t);\,\theta(s)\big) \;=\; \sum_{ij} I\big(\theta_{ij}(t);\,\theta_{ij}(s)\big) \;\le\; \frac{K}{2}\cdot e^{-2|t-s|/\tau_E}\cdot(1+o(1)). \tag{S1.5b}$$

The additivity of MI over independent coordinates is *correct* and circumvents the χ^2-tensorisation trap (multiplicativity $\prod_{ij}(1+\chi^2_{\mathrm{coord}})$ versus the expected sum $\sum_{ij}\chi^2_{\mathrm{coord}}$): MI is additive, χ^2 is not, and so the passage to the final constant $K/2$ is made through MI, not through χ^2.

Step 5d. DPI on ξ and the final MI bound. MI is monotone under measurable maps (Cover and Thomas 2006, ch. 2.8): the coarse-graining of the vector $\theta(\cdot)$ through a bounded measurable ξ does not increase the MI of the pair $(\xi(\theta(t)),\xi(\theta(s)))$ beyond the MI of the original pair $(\theta(t),\theta(s))$. Combining steps 5b–5c: for $|t-s|\ge\tau_E$

$$I\big(\xi(\theta(t));\,\xi(\theta(s))\big) \;\le\; I\big(\theta(t);\,\theta(s)\big) \;\le\; C_2\cdot e^{-2|t-s|/\tau_E}, \tag{S1.5}$$

with explicit constant $C_2 = K/2\cdot(1+o(1))$. The Lipschitz structure of ξ (step S1.4) and the boundedness $|\xi|\le B$ are not used for the MI bound — they are needed only for the constant in the coordinate scale of the covariance form (S1.4'). The conditioning on $M^{\mathrm{refr}}_{\le t}$ is applied through the chain rule on MI: conditioning on $M^{\mathrm{refr}}_{\le t}$ — a function of $\theta(\cdot)$ on $[s_0,t]$ — does not increase the MI of the pair $(\xi(\theta(t)),\xi(\theta(s)))$ beyond (S1.5); the correction of the standard concentration step S1.3 (the trajectory exiting the ball of radius $R(\delta)$) enters the additional term $\delta\cdot\ln k$ of the final statement (S1.6).

Alternative chain. The same decay is obtained through the Mehler semigroup representation of OU and the spectral gap of the ergodic diffusion: for the stationary OU measure and bounded measurable ξ the correlations and MI decay by the standard result on the spectral gap of Gaussian semigroups (Bakry–Émery functional inequality; see (Pavliotis 2014, ch. 4) on the spectral decomposition of OU and exponential ergodicity; alternatively, the survey in (Levin et al. 2017, ch. 13) for the discrete analogue of mixing times). This formulation is shorter but requires a reference apparatus beyond § 5.2; the chain through MI-tensorisation (steps 5b–5c) is formally self-contained within the present work.

1.6 S1.6 Final statement

Lemma S1 (supplementary, softmax-regularity of OU). *In the class of § 5.2 (OU parametrisation of the logits $\theta(t)\in\mathbb{R}^K$ with characteristic time $\tau_E = 1/\lambda$ and stationary variance $\sigma^2/(2\lambda)$), for any bounded measurable functional $\xi:\mathbb{R}^K\to\mathbb{R}$ with $|\xi|\le B$ and coordinate-wise Lipschitz constant L_ξ in the ℓ_∞ metric, and for any $\delta\in(0,1)$, there exists a constant $C = C(\delta,B,L_\xi,\sigma,\lambda,k)$ such that for all s,t with $|t-s|\ge\tau_E$ one has*

$$I\big(\xi(\theta(t));\, \xi(\theta(s)) \;\big|\; M^{\mathrm{refr}}_{\le t}\big) \;\le\; C \cdot e^{-2|t-s|/\tau_E} + \delta \cdot \ln k. \tag{S1.6}$$

In particular, for the projections $P_{ij}(\theta(\cdot))$ of the row-wise softmax parametrisation of the transition matrix one has $L_\xi \le 1$ and $B \le 1$, which gives an explicit constant C independent of k for fixed σ^2/λ.

Lemma S1 formally closes the step in the proof of Lemma 2 (§ 4.4) that requires the vanishing of $I(M^{\mathrm{stale}}; X_E \mid M^{\mathrm{refr}})$ for $\Delta t \gg \tau_E$: the right-hand side of (S1.6) is summed over the age differences in the FIFO distribution and yields a total contribution of the stale fraction of order $O(c)$, which is what is used in the main proof of Lemma 2 upon substitution into (8a) through (7). The condition $|t-s| \ge \tau_E$ is the characteristic scale on which the exponential is not degenerate; for $|t-s| < \tau_E$ the MI is bounded by $H(\xi) \le \ln k$ trivially, which fits into the additional constant term.

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2 S2. Full proof of Remark 5 (per-bit-uniformity for OU)

Remark 5 (§ 4.4) states the additivity of MI over the independently refreshed coordinates of the logits of the OU parametrisation of § 5.2 with constant $1 + o(1)$ — the assumption of Lemma 2 that provides the passage from the arithmetic fraction of refreshed bits c to the information-theoretic bound $\nu \ge 1 - c$ in the slow-driving limit. Proof: (i) OU as K independent processes; (ii) Gaussian posterior on each coordinate; (iii) exact additivity of MI over coordinates; (iv) FIFO in the K-coordinate decomposition; (v) comparison with the optimal observer; (vi) the final bound with $\kappa = 1 + o(1)$; (vii) limits under violation of independence.

2.1 S2.1 OU as K independent processes on the coordinates

In § 5.2 / § 6.2 the OU parametrisation defines the logits $\theta_{ij}(t)$ ($i \ne j$, $K = k(k-1)$ free parameters) as independent one-dimensional processes:

$$d\theta_{ij} = -\lambda(\theta_{ij} - \theta^*_{ij})\, dt + \sigma\, dW_{ij}, \tag{S2.1}$$

where W_{ij} are independent Wiener processes, $\theta^*_{ij} = 0$ by the convention of § 6.2. The diagonal logits $\theta_{ii} \equiv 0$ are fixed as the anchor of the softmax normalisation. The joint distribution $\theta(t) = (\theta_{ij}(t))_{ij}$ in the stationary regime is Gaussian with diagonal covariance $\Sigma = (\sigma^2/(2\lambda))I_K$: the coordinates are statistically independent.

2.2 S2.2 Gaussian posterior on each coordinate

Upon observing $X_E^{[0,t]}$ under the true $P(t) = \mathrm{softmax}(\theta(t))$ the learner's posterior over $\theta(t)$, under regular parametric estimation (MLE or the Robbins–Monro approximation to the Bayesian update of § 6.2), is asymptotically normal with covariance $\Sigma_{\mathrm{post}}(t)$, whose entries decay as $O(1/t)$ — the standard result on the asymptotic normality of maximum-likelihood estimates for regular families (Clarke and Barron 1990, § 3).

Block structure of the Fisher information. The information matrix $\mathcal{F}(\theta)$ for OU with independent coordinates and a row-wise softmax is block-diagonal *by rows* i: different rows give an independent contribution to the likelihood through the independent samples $\{X_t \to X_{t+1}\}$ when $X_t = i$. The additivity by rows is exact, not approximate.

Within-row coupling. Within a row i the Fisher block has the form $\mathcal{F}^{(i)}(\theta) = n_i(t) \cdot (\mathrm{diag}(p_i) - p_i p_i^T)$, where $p_i = (P_{ij})_{j=1}^k$ is the i-th row of the softmax and $n_i(t)$ is the number of visits to state i by time t. This softmax-Fisher matrix is **not diagonal**: the softmax normalisation couples all j through the common denominator. The exact additivity of the diagonal approximation $\Sigma_{\mathrm{post}} \approx \mathrm{diag}(\sigma_{ij}^2)$ therefore requires justification.

Spectral analysis via the Schur formula. The matrix $\mathrm{diag}(p) - pp^T$ is the standard centring projector: it has eigenvalue 0 in the direction $\mathbf{1}$ (which corresponds to the constraint $\sum_j p_{ij} = 1$, operationally — the fixing of $\theta_{ii} \equiv 0$ as the softmax anchor, see S2.1) and eigenvalues $\{p_{ij}\}_{j \neq i}$ on the orthogonal subspace of free parameters. The condition number of the Fisher block in the free coordinates:

$$\kappa(\mathcal{F}^{(i)}_{\mathrm{free}}) \;=\; \frac{\max_{j \neq i} p_{ij}}{\min_{j \neq i} p_{ij}}. \tag{S2.2a}$$

On the event of step S1.3 (the OU trajectory in the ball $R(\delta)$ with probability $\geq 1 - \delta$), for softmax one has $\max_j p_{ij} / \min_j p_{ij} \leq e^{2R(\delta)}$, whence $\kappa(\mathcal{F}^{(i)}_{\mathrm{free}}) \leq e^{2R(\delta)}$ — bounded and dependent only on the radius of the OU ball through $R(\delta) = \sigma\sqrt{\ln(2K/\delta)/\lambda}$ (S1.3).

Correction to the diagonal approximation. The Gaussian approximation of the posterior gives $\Sigma^{(i)}_{\mathrm{post}} = (\mathcal{F}^{(i)}_{\mathrm{free}})^{-1}$ with the same spectrum of inverse eigenvalues; the off-diagonality of Σ_{post} within a row generates a correction to the exact additivity (S2.2). In the slow-driving limit $\lambda\tau_{\mathrm{relax}} \ll 1$ the correction is of order $O(\kappa(\Sigma^{(i)}_{\mathrm{post}}) \cdot \lambda\tau_{\mathrm{relax}})$: the condition number enters multiplicatively, since the off-diagonal contribution to the MI is bounded through the trace of $\Sigma^{(i)}_{\mathrm{post}}$, majorised by κ times the diagonal part. On the OU ball S1.3 this gives

$$\Sigma_{\mathrm{post}} \;=\; \mathrm{diag}(\sigma_{ij}^2(t)) \cdot \big(1 + O\big(e^{2R(\delta)} \cdot \lambda\tau_{\mathrm{relax}}\big)\big), \tag{S2.2b}$$

which remains $1 + o(1)$ under the strengthened slow-driving condition $e^{2R(\delta)} \cdot \lambda\tau_{\mathrm{relax}} \ll 1$. In the numerical regimes of § 6.2 ($\sigma = 0.1$, $\lambda = 10^{-3}$, $K = 56$, $\delta = 0.05$) one obtains $R(\delta) \approx 0.1 \cdot \sqrt{\ln(2240)/10^{-3}} \approx 8.8$ — too large for the literal $e^{2R(\delta)} \cdot \lambda\tau_{\mathrm{relax}} \ll 1$; however, (S2.2b) is a *uniform* upper estimate on the $R(\delta)$-ball, whereas the typical softmax normalisation is concentrated near equilibrium and $\kappa(\mathcal{F}^{(i)}_{\mathrm{free}})$ is empirically close to 1 (the numerical control of § 6.2 shows $1+o(1)$ on average over realisations). An analytical improvement of the estimate (S2.2b) — through the concentration of softmax around the typical logit, rather than the uniform bound — is left as a refinement in § 8.5.

2.3 S2.3 Additivity of MI over independent coordinates

For a Gaussian posterior $\hat{\theta} = (\hat{\theta}_{ij})$ with diagonal covariance $\Sigma_{\text{post}} = \text{diag}(\sigma_{ij}^2)$, the *exact* additivity of MI with the true latent holds:

$$I(\hat{\theta};\, \theta) \;=\; \sum_{ij} I(\hat{\theta}_{ij};\, \theta_{ij}). \tag{S2.2}$$

Derivation: by the chain rule (Cover and Thomas 2006, ch. 2.5)

$$I(\hat{\theta};\, \theta) \;=\; \sum_{ij} I\Big(\hat{\theta}_{ij};\, \theta_{ij} \,\Big|\, \hat{\theta}_{<ij},\, \theta_{<ij}\Big)\,;$$

the independence of the OU coordinates gives $\theta_{ij} \perp (\theta_{<ij})$ (statistical independence by construction), and the diagonality of Σ_{post} gives $\hat{\theta}_{ij} \perp \hat{\theta}_{<ij}$ (conditional independence of the posterior estimates given the true latent). The conditional MIs reduce to the unconditional $I(\hat{\theta}_{ij}; \theta_{ij})$, which yields (S2.2).

The correction to the exact additivity for $\lambda\tau_{\text{relax}} = O(\epsilon)$ is a term $O(\epsilon)$, absorbed into $1 + o(1)$ as $\epsilon \to 0$.

2.4 S2.4 Refresh-fraction FIFO in the *K*-coordinate decomposition

In the FIFO memory-update scheme of Lemma 2, a refresh fraction of size c refreshes a fraction c of the K coordinates over an interval τ_E uniformly (the definition of FIFO and the formulas $(*)$). By the additivity (S2.2) the MI of the refreshed fraction with the true latent of the environment decomposes:

$$I(M_{\le t}^{\text{refr}};\, X_E^{[t,t+\tau]}) \;=\; \sum_{ij\in\text{refr}} I(\hat{\theta}_{ij}^{\text{refr}};\, \theta_{ij}(t+\tau)) \;=\; c\cdot K\cdot \bar{I}_{\text{coord}}(t+\tau), \tag{S2.3}$$

where $\bar{I}_{\text{coord}}(t+\tau) := K^{-1}\sum_{ij} I(\hat{\theta}_{ij}; \theta_{ij}(t+\tau))$ is the average MI per coordinate under the ideal (Bayes-optimal for a regular family) estimate. Structurally: each refreshed coordinate contributes independently, proportionally to $\bar{I}_{\text{coord}}$; the total MI is proportional to the fraction of refreshed coordinates c under K-fold additivity.

2.5 S2.5 Comparison with the optimal observer

The MI of the optimal observer (the full sufficient statistic over $\theta(t)$) on the horizon τ in the slow-driving limit:

$$I_{\text{pred}}^{\text{opt}}(t,\tau) \;=\; I(X_t;\, X_{t+\tau} \mid \theta(t)) \;=\; \sum_{ij} I_{\text{coord}}(t+\tau) \;=\; K\cdot \bar{I}_{\text{coord}}(t+\tau), \tag{S2.4}$$

where the decomposition over coordinates uses the conditional independence of the transitions $X_t \to X_{t+\tau}$ over different rows and the independence of the logit coordinates θ within a single row in the slow-driving limit. The approximation $\theta(t+\tau) \approx \theta(t)$ works to accuracy $O(\sigma^2\tau/\lambda)$: over the horizon τ, when $\lambda\tau \ll 1$, the latent drift is small, and the MI per coordinate $I_{\mathrm{coord}}(t+\tau) = I_{\mathrm{coord}}(t) \cdot (1 + O(\lambda\tau))$.

2.6 S2.6 The final bound

Substituting (S2.3) into (S2.4):

$$\frac{I(M^{\mathrm{refr}}_{\le t};\, X_E^{[t,t+\tau]})}{I^{\mathrm{opt}}_{\mathrm{pred}}(t,\tau)} \;=\; \frac{c \cdot K \cdot \bar{I}_{\mathrm{coord}}(t+\tau)}{K \cdot \bar{I}_{\mathrm{coord}}(t+\tau)} \;=\; c \cdot (1 + O(\lambda\tau)).$$

That is,

$$I(M^{\mathrm{refr}}_{\le t};\, X_E^{[t,t+\tau]}) \;\le\; c \cdot I^{\mathrm{opt}}_{\mathrm{pred}}(t,\tau) \cdot (1 + O(\lambda\tau)). \tag{S2.5}$$

For $\lambda\tau \ll 1$ (the slow-driving regime, satisfied by the condition of the class of § 5.2) and under the strengthened condition $e^{2R(\delta)} \cdot \lambda\tau_{\mathrm{relax}} \ll 1$ from S2.2 (within-row Fisher coupling), $1 + O(\lambda\tau) + O(e^{2R(\delta)}\lambda\tau_{\mathrm{relax}}) = 1 + o(1)$, and (S2.5) gives per-bit-uniformity with constant $\kappa = 1 + o(1)$. Without the strengthened condition, $\kappa = 1 + O(e^{2R(\delta)}\lambda\tau_{\mathrm{relax}})$, which for finite δ, σ, λ gives a multiplicative correction but does not cancel the qualitative structure of (8a) (see also S2.7 for additional sources of non-diagonality). Substitution into (7) under the condition $I(M^{\mathrm{stale}}; X_E \mid M^{\mathrm{refr}}) \to 0$ (Lemma S1) gives asymptotically

$$\nu^{\mathrm{theor}}(t) \;=\; 1 - \frac{I(M^{\mathrm{refr}}_{\le t};\, X_E^{[t,t+\tau]})}{I^{\mathrm{opt}}_{\mathrm{pred}}(t,\tau)} \;\ge\; 1 - c \cdot (1 + o(1)), \tag{S2.6}$$

which is the statement (8a) of Lemma 2 with explicit constant $\kappa = 1 + o(1)$.

2.7 S2.7 Discussion of limits: correlated coordinates

Under violation of the independence of the OU coordinates (cross-arrow drift with non-diagonal Σ in (S2.1)) per-bit-uniformity requires refinement. Besides the between-row *prior* correlation, the second source of non-diagonality is the within-row Fisher coupling through the softmax normalisation (S2.2): the condition number of the Fisher block $\kappa(\mathcal{F}^{(i)}_{\mathrm{free}}) = \max_j p_{ij} / \min_j p_{ij}$ controls the deviation from exact additivity (S2.2a, S2.2b) and enters multiplicatively into the final correction (S2.6). Both sources are combined in a common condition-number argument.

Let $\kappa(\Sigma_{\mathrm{post}})$ be the condition number (the ratio of the maximal to the minimal eigenvalue) of the joint covariance of the posterior, combining the between-row prior correlations (S2.1, under non-diagonal Σ_{prior}) and the within-row Fisher correlations (S2.2). Through the Gaussian representation of MI $I(\hat{\theta}; \theta) = (1/2)\ln\det(\Sigma_{\mathrm{prior}}\Sigma^{-1}_{\mathrm{post}}) = (1/2)\ln\det(I + \mathcal{F}\,\Sigma_{\mathrm{prior}})$ (where $\mathcal{F} = \Sigma^{-1}_{\mathrm{post}} - \Sigma^{-1}_{\mathrm{prior}}$ is the observed Fisher information, and $\Sigma^{-1}_{\mathrm{post}} = \Sigma^{-1}_{\mathrm{prior}} + \mathcal{F}$) and the Hadamard inequality $\det(\Sigma) \le \prod_i \Sigma_{ii}$ one obtains an upper bound that in the worst case is majorised by a value $\kappa(\Sigma_{\mathrm{post}})$ times larger than the additive sum. In terms of the constant of Remark 5:

$$\kappa \;\le\; \kappa(\Sigma_{\mathrm{post}}) \cdot (1 + o(1)). \tag{S2.7}$$

For a diagonal Σ_{post} (independent coordinates) $\kappa(\Sigma_{\mathrm{post}}) = 1$ and (S2.7) reduces to $\kappa = 1 + o(1)$. For correlated coordinates the bound is weakened multiplicatively by $\kappa(\Sigma_{\mathrm{post}})$; a numerical estimate for a specific cross-parametrisation is the subject of a separate study. In the OU parametrisation of § 5.2 with independent W_{ij} and a diagonal anchor $\theta_{ii} \equiv 0$, the condition $\kappa(\Sigma_{\mathrm{post}}) = 1$ holds by construction, and per-bit-uniformity with $\kappa = 1 + o(1)$ applies without restrictions.

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3 S3. Extended notation table

A glossary of the notation of the work indicating the dimension and the defining equation; “dimensionless” denotes a dimensionless quantity. Abbreviations: “MI” — mutual information (Cover and Thomas 2006, ch. 2); “TV” — total variation; “KL” — Kullback–Leibler. The contexts of definition are § X.Y and equations (N) of the main text.

3.1 S3.1 Efficiency and information flow

Symbol	Dimension/type	Definition	Theoretical upper bound
$\eta_L(t)$	dimensionless, $\in [0,1]$	Non-stationary Landauer efficiency; (10) § 5.1	1 (irreversible regime, asymptotics $t \gg \tau_d$)
$\eta_L^{(\tau_w)}(t)$	dimensionless, $\in [0,1]$	Efficiency with sliding window τ_w; (10a) § 5.2	1 (irreversible regime)
η_L^{stat}	dimensionless	Stationary limit of η_L; § 5.3, (Andriishin 2026, § 2.1)	1 (irreversible regime)
$\eta_L^{\mathrm{excess}}(t)$	dimensionless	Secondary efficiency through L_{excess}; (S8.3) § S8	no universal upper bound (secondary metric)
$I_{\mathrm{pred}}(t,\tau)$	bits	One-step predictive information at $\hat{P}(t)$; (4) § 3.1	$\ln k$ (one-step, $k = \lvert\Omega\rvert$)
$I_{\mathrm{pred}}^{\mathrm{stat}}(\tau)$	bits	Stationary limit of I_{pred}; § 2.2	$\ln k$ (one-step); $\lvert M_t\rvert \log 2$ (capacity)
$I_{\mathrm{pred}}^{\mathrm{opt}}(t,\tau)$	bits	MI of the ideal observer with access to $\theta(t)$; (7) § 4.1	$\ln k$ (one-step, $X_t, X_{t+\tau}$ discrete)
$I_{\mathrm{pred}}^{\mathrm{Bialek}}(X_E,\tau)$	bits	Classical predictive information (Bialek et al. 2001); § 2.2 (Andriishin 2026)	C_μ (excess entropy, Crutchfield and Feldman 2003)
$N_{\max}(t)$	bits	Cumulative Landauer budget; (1) § 2.1	$\lvert M_t\rvert \log 2$ per active part of memory; unbounded above in time
$N_{\max}^{(\tau_w)}(t)$	bits	Budget on the sliding window τ_w; (10a) § 5.2	$\lvert M_t\rvert \log 2$ per active part
$L_{\mathrm{excess}}(t)$	nats	Cumulative excess-loss of the learner; (S8.1) § S8	asymptotics $(K/2)\ln(\lambda t)$ — conjecture (S8.2)

Distinction of three types of theoretical bounds. (i) The one-step bound $\ln k$ — the standard capacity of a discrete channel with alphabet Ω (Cover and Thomas 2006); it controls the instantaneous information capacity through one step of the environment. (ii) The capacity bound $|M_t| \log 2$ — the structural ceiling for the integral information that a finite memory M_t of $|M_t|$ bits can store. (iii) The excess entropy C_μ (Crutchfield and Feldman 2003) — the total predictive information between the past and the future of a stationary process, in general infinite for nontrivial classes. Distinction: $\ln k$ — for the one-step $I_{\text{pred}}(t, \tau)$, $|M_t| \log 2$ — for capacity estimates of memory, C_μ — for interval-MI of the type $I_{\text{pred}}^{\text{Bialek}}$.

3.2 S3.2 Energy flows

Symbol	Dimension/type	Definition
$E_{\text{actual}}^{\text{curr}}(t)$	J	Current operational dissipation; § 2.1
$E_{\text{store}}(t)$	J	Cumulative cost of maintaining old memory; § 2.1
$E_{\text{grow}}(t)$	J	Cumulative cost of memory expansion; § 2.1
$\dot{E}_{\text{store}}(t)$	W	Instantaneous maintenance power; Lemma 1 § 2.3
$\dot{E}_{\text{grow}}(t)$	W	Instantaneous expansion power; § 2.4
$\dot{E}_{\text{store}}^{\text{nostalg}}(t)$	W	Maintenance power of the nostalgic fraction; (8) § 4.2
$E_{\text{store}}^{(1)}(t)$	J	Minimal dissipation for maintaining one bit; (2) Lemma 1
$E_{\text{grow}}^{(1)}$	J	Minimal dissipation for creating one bit of capacity; (3) § 2.4
η_{ex}	dimensionless	Exergetic efficiency of metabolism; (Andriishin 2026, § 3.5)

3.3 S3.3 Memory and its layers

Symbol	Dimension/type	Definition
M_t	bits (size); configuration	Current (refreshable) memory layer; § 3.1
$M_{<t}$	bits; configuration	Archival memory layer; § 3.1
$M_{\le t} = (M_t, M_{<t})$	bits; configuration	Full memory of the system; § 3.1
$M_{\le t}^{\mathrm{refr}}$	bits	Refresh fraction (refreshed over the interval τ_E); § 4.4
$M_{\le t}^{\mathrm{stale}}$	bits	Stale fraction (not refreshed over τ_E); § 4.4
$\lvert M_{\le t}\rvert$	bits	Size of the active part of memory; § 4.4, (11b) § 5.3
$\lvert M_{\le t}^{\mathrm{cap}}\rvert$	bits	Structural limit of the memory capacity; (12) § 5.3
$\dot{M}_{\mathrm{refresh}}(t)$	bits/s	Memory refresh rate; § 4.4, (13) § 7
$\dot{M}_{\mathrm{grow}}(t)$	bits/s	Memory capacity growth rate; (13) § 7
$\dot{M}_{\mathrm{refresh}}^{\mathrm{eff}}(t)$	bits/s	Effective refresh rate for Robbins–Monro; § 6.2

3.4 S3.4 Latent parameters and estimates

Symbol	Dimension/type	Definition
$\theta(t)$	$\mathbb{R}^K$	True logits of the environment's transition matrix; § 5.2
$\hat{\theta}(t)$	$\mathbb{R}^K$	Learner's estimate of the logits; § 6.2
$\theta_{ij}(t)$	$\mathbb{R}$	Logit coordinate, $i \neq j$; § 5.2
$P(t)$	$k \times k$ stochastic	Environment's transition matrix; § 5.2
$\hat{P}(t)$	$k \times k$ stochastic	Learner's estimate of the transition matrix; § 3.1
X_t	$\in \Omega$	State of the environment at time t; § 3.1
$X_E^{[t,t+\tau]}$	trajectory	Realisation of the environment on the window $[t, t+\tau]$; (1a) (Andriishin 2026)

3.5 S3.5 Model parameters and time scales

Symbol	Dimension/type	Definition
k	integer; $\|\Omega\|$	Size of the environment's state alphabet; § 5.2, § 6.1
$K = k(k-1)$	integer	Dimension of the logits of the OU parametrisation; § 5.2
Ω	finite set	State alphabet of the Markov chain; § 3.1
λ	1/s	OU relaxation rate; the inverse of the characteristic drift time τ_E; § 5.2
σ	$\sqrt{1/\mathrm{s}}$	Amplitude of the OU noise; § 5.2
α	dimensionless	Dirichlet parameter for the PSP; § 6.1
τ	s	Prediction horizon of the one-step I_{pred}; (4) § 3.1
τ_d	s	Characteristic dissipation/response time of the system; § 2.1 (Andriishin 2026)
τ_w	s	Width of the estimator's sliding window; § 6.1, § 5.2
$\tau_E = 1/\lambda$	s	Characteristic drift time of the environment; § 5.2
τ_{relax}	s	Relaxation time of the Markov chain at a slice; § 5.2, § 6.1
$\tau_{1/2}$	s	Half-life of the bit-read accuracy; Lemma 1 § 2.3
τ^*_{reset}	s	Optimal memory reset interval; (12) § 5.3
T	K	Temperature of the receiving heat bath; § 2.1 (Andriishin 2026)
ε	dimensionless, $\in (0, 1/2)$	Accuracy threshold of bit reading; Lemma 1 § 2.3
γ	1/s	Symmetric transition rate in Lemma 1; § 2.3
η_t	dimensionless	Robbins–Monro step of the learner; § 6.2

3.6 S3.6 Nostalgia and the regime threshold

Symbol	Dimension/type	Definition
$\nu(t)$	dimensionless, $\in [0, 1]$	Nostalgia (general symbol, the context determines the variant); § 4.1
$\nu^{\text{theor}}(t)$	dimensionless, $\in [0, 1]$	Theoretical nostalgia through $I_{\text{pred}}^{\text{opt}}$; (7) § 4.1
$\nu^{\text{op}}(t)$	dimensionless, $\in [0, 1]$	Operational nostalgia through C_{μ}^{emp}; (7') § 4.1
ν_c	dimensionless, $\in (0, 1)$	Critical value / regime-transition threshold; (11b) § 5.3
ν_c^{theor}	dimensionless, $\in (0, 1)$	Theoretical critical nostalgia; (11b) § 5.3
ν_c^{op}	dimensionless, $\in (0, 1)$	Operational critical nostalgia; § 8.6 Prediction 1
ν_c^{emp}	dimensionless, $\in (0, 1)$	Empirical proxy from § 6.1 (PSP simulation); § 6.1
c	dimensionless	Computable constant $\dot{M}_{\text{refresh}}\tau_E/\lvert M_{\leq t}\rvert$; (*) § 4.4
η_M	dimensionless	Memory-balance ratio $\dot{M}_{\text{refresh}}\tau_{1/2}/\lvert M_{\leq t}\rvert$; (11b) § 5.3
κ	dimensionless, ≥ 1	Per-bit-uniformity constant of Remark 5; § 4.4, § S2.6
$\rho(t)$	dimensionless	Informational rejuvenation metric; (13) § 7

3.7 S3.7 Information complexity and diagnostic proxies

Symbol	Dimension/type	Definition
$C_v^{\text{static}}(t)$	bits	Structural complexity (Lempel–Ziv, assembly); § 4.3
$C_v^{\text{predictive}}(t)$	bits	Dynamic predictive power; (9) § 4.3
C_μ	bits	Statistical complexity of Crutchfield–Feldman (Crutchfield and Feldman 2003)
$C_\mu^{\text{emp}}(t, \tau)$	bits	Empirical excess entropy on a sliding window; (7') § 4.1
h_μ	bits/symbol	Entropy rate; § 4.3

3.8 S3.8 Active inference (§ 7)

Symbol	Dimension/type	Definition
$G[\pi]$	nats	Expected free energy of policy π; § 7 / S7
$G^{\mathrm{epist}}[\pi]$	nats	Epistemic component of EFE (information gain); § 7 / S7
$G^{\mathrm{prag}}[\pi]$	nats	Pragmatic component of EFE (preference); § 7 / S7
$q(s, o, \pi)$	density	Generative model in active inference (Friston et al. 2017a; Parr et al. 2022); § 7 / S7
$p(o', s' \mid C)$	density	Preference distribution with target parameter C; § 7 / S7
$\beta = 1/(k_B T)$	1/J	Inverse temperature for converting nats to J; (S7.1) § 7 / S7

3.9 S3.9 Fundamental constants

Symbol	Dimension	Value / definition
k_B	J/K	Boltzmann constant
$k_B T \ln 2$	J	Minimal payment for one erasure (Landauer 1961)
$\ln 2$	dimensionless	Conversion factor nats → bits

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4 S4. Simulation parameters for reproducibility

This section assembles the technical parameters of the numerical illustrations of § 6.1 (PSP), § 6.2 (OU) and § S8.3 (adiabatic scan). The implementation is `andriishin/landauer-nostalgia-oa`, directories `simulations/markov_drift{,_ou,_ou_iinf_adiab}/`. The seed SEED = 20260524 with per-run offsets; running `python main.py` in each directory reproduces the cited numerical values up to the version of the libraries used.

4.1 S4.1 PSP simulation (§ 6.1)

Model class. A piecewise-stationary Markov process with Poisson switches of the transition matrix. Alphabet $\Omega = \{1, \ldots, k\}$, $k = 8$. On the interval $[T_i, T_{i+1})$ the transition matrix $P^{(i)}$ is constant; the switch times $\{T_i\}$ form a Poisson stream with intensity λ. At time T_i the matrix is updated to a random stochastic $P^{(i)}$ with independent rows, each row a Dirichlet with parameter $\alpha = 0.3$.

Base parameters. $k = 8$, $\alpha = 0.3$, $\lambda = 10^{-3}$ (for the drift regimes), prediction horizon $\tau = 1$ step.

Relaxation rate. $\langle \tau_{\mathrm{relax}} \rangle \approx$ 2–3 steps for $\alpha = 0.3$; $\lambda \tau_{\mathrm{relax}} \approx 3 \cdot 10^{-3} \ll 1$ — slow-driving is satisfied with margin.

Three regimes (§ 5.3):

Regime	λ	τ_w	T	Realisations
Stationary	0	100	10^4	80
Drift without reset	10^{-3}	∞	10^5	60
Drift with reset	10^{-3}	$\{100, 500, 2000\}$	10^5	60

Grid of τ_w for Fig. 2. Eight points $\tau_w \in \{50, 100, 200, 500, 10^3, 2 \cdot 10^3, 5 \cdot 10^3, 10^4\}$, $\lambda = 10^{-3}$, $T = 5 \cdot 10^4$, 40 realisations per point.

MI estimator. KL-detuned proxy $\widehat{I}_{\text{pred}}(t, \tau) = I_{\text{opt}}(P(t), \tau) - \mathbb{E}_{X_t \sim \pi(t)}[D_{KL}(P(t)^\tau(\cdot \mid X_t) \| \hat{P}(t)^\tau(\cdot \mid X_t))]$ with clipping at zero from below; cross-checked against independent estimates: the discrete plug-in with Treves–Panzeri correction (Treves and Panzeri 1995) and the continuous KSG estimator with k-NN, $k_{NN} = 3$ (Kraskov et al. 2004); consistency on the grid verified.

Budget N_{max}. In the simulation the normalisation $k_B T \ln 2 = 1$ is adopted and $N_{\max}(t) = N_0 + R \cdot t$ with $R = 1$ unit per step (the base operational payment plus the Lemma 1 payment on $|M_t| = k^2 = 64$ maintained parameters).

Computational cost. Complexity per realisation $O(T \cdot k^2 \cdot \tau_w)$; the base stationary parametrisation ($T = 10^4$, $\tau_w = 100$, $k = 8$) — $\sim 6 \cdot 10^7$ operations.

Key numerical results. Stationary: $\eta_L \approx 6.1 \cdot 10^{-5}$ for $t > 5 \cdot 10^3$. Drift without reset: $\nu_\infty(t > 60000) \approx 0.999 \pm 0.001$, η_L decays by ~ 3 decades. Grid of τ_w: optimum $\tau_w^* \approx 200$, $\langle \eta_L \rangle \approx 1.3 \cdot 10^{-5}$. Empirical $\nu_c^{\text{emp}} \approx 0.44$ at the half-life of η_L.

Reproduction command and outputs. `python simulations/markov_drift/main.py`; wall-time $\sim$ 5–10 minutes; outputs — `results_summary.{txt,json}`, `run.log`, figures `paper/figs/fig{1,2,3}_*.png`.

4.2 S4.2 OU simulation (§ 6.2)

Model class. A Markov chain with continuous OU drift of the logits of the transition matrix. $k = 8$, $K = k(k-1) = 56$ free parameters. The OU equations $d\theta_{ij} = -\lambda(\theta_{ij} - \theta_{ij}^*)\, dt + \sigma\, dW_{ij}$ with $\theta_{ij}^* = 0$, independent W_{ij} for $i \neq j$, and fixed $\theta_{ii} \equiv 0$ as the softmax anchor. Euler–Maruyama discretisation with $\Delta t = 1$.

Base parameters. $k = 8$, $\lambda = 10^{-3}$, $\sigma = 0.1$, $T = 2 \cdot 10^4$ steps, 50 independent realisations of the OU process; initial logits $\theta(0) \sim \mathcal{N}(0, 0.1^2)$.

Dimensionless parameters. $\sigma^2/\lambda = 10$; stationary variance of the logits $\sigma^2/(2\lambda) = 5$ — a strong fluctuation regime (not OU-concentration). Transient window $1/\sigma^2 = 10^2$ steps.

Approximation of the Bayesian learner. Robbins–Monro online update with decreasing step $\eta_t = c_0/\sqrt{t}$, $c_0 = 3$; at each step $\hat{\theta}_{x_t,j}(t+1) = \hat{\theta}_{x_t,j}(t) + \eta_t \cdot (\mathbb{1}[x_{t+1} = j] - \hat{P}_{x_t,j}(t))$ for $j \neq x_t$. The rate $1/\sqrt{t}$ corresponds to the concentration of the posterior along each of the K directions and is asymptotically Bayes-optimal for regular parametric models (Clarke and Barron 1990, § 3).

Effective refresh rate. $\dot{M}_{\text{refresh}}^{\text{eff}}(t) \sim K c_0/(\tau_E \sqrt{t})$; for $K = 56$, $\tau_E = 10^3$, $t = 2 \cdot 10^4$ it gives ~ 0.17 refreshes per step; the constant $(*)$ $c(t) = \dot{M}_{\text{refresh}}^{\text{eff}} \cdot \tau_E/|M_{\leq t}| \approx K c_0/(|M_{\leq t}|\sqrt{t}) \approx 0.02$ for $|M_{\leq t}| = K$. This justifies the numerical gap $\nu(t > 10^4) \approx$

0.82 against the prediction $\liminf \nu^{\text{theor}} \geq 1 - c \approx 0.98$ (§ 6.2; the finiteness of the observation time and Remark 5 cover the discrepancy ≈ 0.16).

Budget. Two variants: (a) $N_{\max} = Rt$ with $R = 1$ and (b) the full $N_{\max}^{\text{full}} = Rt + \nu(t) \cdot K/\tau_{1/2}$ with $\tau_{1/2} = 100$. The curves $\eta_L(t)$ are qualitatively identical.

Key numerical results. Late-time ($t > 10^4$): $\nu = 0.822 \pm 0.031$; $\langle \eta_L \rangle_R \approx 1.0 \cdot 10^{-5}$, $\langle \eta_L \rangle_{\text{full}} \approx 7.3 \cdot 10^{-6}$. Fit of $\eta_L \cdot t$ vs $\ln(\lambda t)$ on $[10^3, 1.5 \cdot 10^4]$: $n = 71$ points, $R^2 = 0.888$ for $N_{\max} = Rt$; $R^2 = 0.906$ for the full budget.

Reproduction command and outputs. `python simulations/markov_drift_ou/main.py`; wall-time $\sim$ 1 minute; outputs — `results_summary.{txt,json}`, `run.log`, figures `paper/figs/fig{4,5}_ou_*.png`.

4.3 S4.3 Adiabatic scan (§ S8.3)

Model class. The OU parametrisation of § 6.2 unchanged ($k = 8$, $\lambda = 10^{-3}$, $K = 56$, Euler–Maruyama $\Delta t = 1$, Robbins–Monro $\eta_t = c_0/\sqrt{t}$, $c_0 = 3$). The measured quantity is the cumulative excess-loss $L_{\text{excess}}(t)$ by (S8.1) with direct access to the ground truth $\theta(s)$.

Parametric scan. Three values of σ at fixed $\lambda = 10^{-3}$:

σ	σ^2/λ	$1/\sigma^2$	T_{total}	Realisations
0.01	10^{-1}	10^4	$5 \cdot 10^4$	25
0.003	$9 \cdot 10^{-3}$	$1.1 \cdot 10^5$	$5 \cdot 10^4$	25
0.001	10^{-3}	10^6	$2 \cdot 10^5$	25

The simulation time T is chosen within the transient window $1/\sigma^2$ for each σ.

Joint fit. $L_{\text{excess}}(t) = A \cdot t + B \cdot \ln(\lambda t) + C$ on the window $t \in [10^3, 10^4]$ (fixed for direct comparison across σ).

Wide-window negative control. An additional fit with the extended window $[10^3, T_{\text{total}}]$: for $\sigma^2/\lambda = 10^{-3}$ and $T_{\text{total}} = 2 \cdot 10^5$ the coefficient $B/(K/2)$ grows to 2.0 — an artefact of the Robbins–Monro $1/\sqrt{t}$ bias of the learner on long horizons, not part of the BNT effect. This control records the sensitivity of the result to the choice of window and prevents the interpretation of the monotone trend of $B/(K/2)$ as the result of selective presentation.

Key numerical results. $B/(K/2) = \{0.54, 0.78, 0.85\}$ for $\sigma^2/\lambda \in \{10^{-1}, 9 \cdot 10^{-3}, 10^{-3}\}$ respectively; $R^2 \geq 0.9995$ at all points; the drift-share $A \cdot t/L_{\text{excess}}(t)$ at $t = 10^4$ falls from 76% to 10%. The monotone growth of B as $\sigma^2/\lambda \to 0$ is consistent with the conjecture of § S8.1, but does not prove convergence $B \to K/2$; 5–6 scan points are needed with an explicit analytical form of the finite-σ correction $B(\epsilon) = (K/2)(1 - g(\epsilon))$.

Reproduction command and outputs. `python simulations/markov_drift_ou_iinf_adiab/main.py`; wall-time $\sim$ 1 minute; outputs — `results_summary.{txt,json}`, `run.log`, figures `simulations/markov_drift_ou_iinf_adiab/fig_adiab_*.png`, copied into `paper/figs/fig{6,7}_adiab_*.png`.

4.4 S4.4 Reproducibility: dependencies and archiving

Seed and dependencies. The global SEED = 20260524 is set at the start of `main.py` of each script; per-run offsets ensure the independence of realisations. Dependencies — `numpy`, `scipy`, `scikit-learn` (KSG), `matplotlib`; exact versions are in `requirements.txt` of each directory.

Full set of commands. At the root of the repository, in sequence, `python simulations/markov_drift{,_ou,_ou_iinf_adiab}/main.py`; total wall-time on a mid-range laptop $\sim$ 7–12 minutes.

Zenodo archiving. A versioned snapshot has been archived on Zenodo (DOI: 10.5281/zenodo.20653051, release `v1.0-submission`); reproduction is also possible by cloning `andriishin/landauer-nostalgia-oa` at the `v1.0-submission` tag (the same snapshot archived on Zenodo).

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5 S5. The full apparatus of § 5 main: Remarks 1–6 to Lemma 2, the protocol of § 5.2, ν_c and τ^*_{reset}

This section assembles the formal details of § 5 main, moved here to lighten the main exposition. Structure: S5.1 — the six Remarks to Lemma 2 (§ 4.4 main fixes the formulation (8a) and the brief statement; here is their full development); S5.2 — point 5 of the protocol of § 5.2 main (the ban on revising the definition of the object of measurement) with justification through (Wolpert 1996) no-free-lunch; S5.3 — the balance condition for the existence of ν_c through (11b) and a discussion of its structural status; S5.4 — the optimal τ^*_{reset} (12), its connection to dynamic regret bounds (Besbes et al. 2015); S5.5 — the analogy with spin glasses through (Hopfield 1982; Amit et al. 1985; Engel and Van den Broeck 2001).

5.1 S5.1 Six Remarks to Lemma 2

Remark 1 (stationary limit). The stationary limit of Lemma 2 is given *not* by $\tau_E \to \infty$ (this passage violates the condition $c < 1$ of $(*)$ main), but by equation (1) main with $\dot{E}_{\text{store}} = \dot{E}_{\text{grow}} = 0$ (§ 2.2 main): in this regime the nostalgic bits carry no payment, and the scale ν^{theor} through (7) main loses its thermodynamic meaning (likewise for ν^{op} as $\hat{P} \to P_0$). Thus, Lemma 2 does not contradict the stationary lemma (Andriishin 2026, § 2.1), but *supplements* it to a new regime: for $\dot{E}_{\text{store}} > 0$ or $\dot{E}_{\text{grow}} > 0$ the condition (8a) is operative, and in the stationary limit it degenerates through the vanishing of both payments.

Remark 2 (computability and falsifiability of $(*)$). The constant c from $(*)$ main is a *computable* function of three measurable parameters $(\dot{M}_{\text{refresh}}, \tau_E, |M_{\le t}|)$, not a free parameter of the theory. Lemma 2 applies at measured $c < 1$; for $c \geq 1$ the system passes into the *sliding-window* regime: over the interval τ_E the entire memory is fully refreshed, there is no archival layer, and the non-stationary theory degenerates. The test on c is an empirical condition of applicability, separating systems with an archival layer from those without it. Long-context LLMs with an M_t layer of order

10^5–10^6 tokens at inference give $\dot{M}_{\text{refresh}}\tau_E \gg |M_t|$, $c \gg 1$, and Lemma 2 assigns them to the sliding window — the theory by its own condition removes this layer from the domain of applicability (§ 8.3 main).

Remark 3 (substantiveness of (8a) beyond the definition). The fraction $1 - c$ is arithmetic (the ratio of the number of unrefreshed bits to the total memory size), while ν^{theor} by (7) main is information-theoretic (the normalised MI deficit). The substantive part of Lemma 2 is the proof that the MI contribution of an unrefreshed bit to $I_{\text{pred}}(t,\tau)$ vanishes asymptotically by DPI through the decorrelation of the OU latent with characteristic time τ_E (see the detailed proof in S1 supplementary through MI-tensorisation over the independent coordinates of OU, § S1.5). Three conditions are required: (i) decorrelation of the optimal statistic with τ_E; (ii) the ideal observer in the denominator of (7) main; (iii) FIFO refresh. Under violation of any of (i)–(iii) the inequality (8a) does not follow from the arithmetic c.

Remark 4 (extension to the general non-stationary class). The basic Lemma 2 is proven for the OU parametrisation of the logits (§ 5.2 main, S1–S2 supplementary). The PSP surrogate (§ 6.1 main) and multiscale drift reproduce the result *illustratively* for $\lambda\tau_{\text{relax}} \ll 1$ through a single decorrelation mechanism with effective τ_E, but there is no rigorous proof for non-OU classes in the present work; see § 8.5 main, open problem. The Wolpert no-free-lunch (Wolpert 1996) gives a fundamental justification for narrowing the class of § 5.2: a universal theory of non-stationary learnability without an a priori restriction of the class is impossible; the slow-driving class is a necessary narrowing, not a methodological choice.

Remark 5 (per-bit-uniformity). The passage from the arithmetic fraction of unrefreshed bits c to the information-theoretic inequality $\nu \geq 1 - c$ requires the additivity of MI over independently refreshed bits:

$$I(M^{\text{refr}}; X_E^{[t,t+\tau]}) \leq c \cdot I_{\text{pred}}^{\text{opt}} \cdot (1 + o(1)).$$

In the slow-driving limit the refresh fraction accumulates no cross-bit redundancy (Clarke and Barron 1990); the full proof of this statement for the OU parametrisation is S2 supplementary. Under violation of the condition $c \to \kappa c$, $\kappa \geq 1$, and Lemma 2 is formulated with a multiplicative correction $\nu \geq 1 - \kappa c$; in the OU class with independent W_{ij} and a diagonal anchor $\theta_{ii} \equiv 0$ the per-bit-uniformity condition is satisfied by construction with $\kappa = 1 + o(1)$ (S2.6 supplementary).

Remark 6 (softmax-regularity for OU). The transfer of OU decorrelation from the autocovariance of the logits $\theta(t)$ to any bounded measurable functional $\xi(\theta)$ (in particular, to the entries of the transition matrix $P_{ij}(\theta)$ and to any sufficient statistic of prediction) is ensured by the Lipschitz continuity of softmax in total variation on the set of logits with finite stationary variance $\sigma^2/(2\lambda)$. Through MI-tensorisation over the independent coordinates of OU (§ S1.5) and DPI the exponential decay $\mathbb{E}[\theta(t)\theta(s)] \propto e^{-(t-s)/\tau_E}$ is transferred to any bounded ξ with the same characteristic time τ_E. The full proof is S1 supplementary (five steps: Lipschitz softmax in ℓ_∞; transfer to the TV of the transition matrix; boundedness of the OU trajectory on the ball $R(\delta)$ with probability $1 - \delta$; transfer of the decay to the covariance of ξ; translation of the TV bound into a bound on the conditional MI through χ^2 and DPI).

Lemma 2 in light of Remarks 1–6 is constructive: under $(*)$ it indicates an explicit mechanism of the emergence of nostalgia (the desynchronisation of the optimal feature through OU decorrelation) and a physical obstacle to its vanishing (the finite memory-refresh rate under FIFO scheduling). The joint action of Remarks 5–6 is the structural justification of the claimed asymptotics $\nu \geq 1 - c$ in the OU class of § 5.2 main; Remark 2 fixes the domain of applicability through the measurable condition $c < 1$.

5.2 S5.2 The protocol of § 5.2 main, point 5: the ban on revising the definition of the object of measurement

Point 5 of the protocol (§ 5.2 main) is a pre-registration commitment that protects falsifiability: the operational definition of the measured quantity is fixed *before* the data and is not revised, under the same theory name, after the data are in. When prediction and measurement disagree systematically, two honest moves remain: (a) refute the theory; (b) explicitly change its name, leaving the old name attached to the original definition. The rule is symmetric — it binds the theorist and the empiricist equally.

Why it matters. The standard response to a failed empirical test is either to reject the theory or to narrow its conditions of applicability (add caveats); both are legitimate and keep the test severe. Point 5 closes only a third, methodologically dubious route — revising the *operational definition* of the object of measurement while keeping the theory's name (a classical ad hoc rescue in Popper's sense, disguised as a technical refinement of operationalisation). The symmetry keeps the discipline fair in both directions: the empiricist may not "refine" the definition of the measured quantity post hoc so that the data agree with the theory; the theorist may not "refine" the definition of the predicted quantity post hoc so that they disagree.

How the present work honours it. The cumulative excess-loss $L_{\text{excess}}(t)$ (S8.1) is kept in § S8 as a separate, explicitly labelled counterfactual diagnostic rather than used as the operationally measurable I_{pred}: L_{excess} requires access to the ground truth $\theta(s)$, unavailable in real biological systems, so conflating it with the one-step MI (4) main under the name "predictive information" would be a post hoc revision of the object of measurement. The theoretical constructions (Lemma 2, ν_c, $C_v^{\text{predictive}}$) rest on ν^{theor} by (7) main; the operational predictions (§ 8.6 main, Prediction 1, Prediction 2) rest on ν^{op} by (7') main. Substitution under one name is excluded; separated accounting of the two quantities is not.

5.3 S5.3 Existence of ν_c: the balance condition (11b) and its structural status

The balance estimate from the condition of equality of the increment of $\dot{I}_{\text{pred}}$ and the losses from staleness (equation (11b) main):

$$\nu_c^{\text{theor}} \cdot |M_{\leq t}|/\tau_{1/2} = (1 - \nu_c^{\text{theor}}) \cdot \dot{M}_{\text{refresh}}.$$

The dimensionless $\eta_M := \dot{M}_{\text{refresh}}\tau_{1/2}/|M_{\leq t}|$ formally gives $\nu_c^{\text{theor}} = \eta_M/(1+\eta_M)$; direct substitution in this simplest form gives numerical values inconsistent with the empirical proxy of § 6.1 main ($\nu_c^{\mathrm{emp}} \approx 0.44$ vs. $\nu_c \approx 0.016$ for the η_M of the base parametrisation of the PSP — a discrepancy of more than an order of magnitude).

Structural status of (11b). (11b) is treated as a *structural balance condition*, not a quantitative predictor: the work postulates the existence of $\nu_c^{\mathrm{theor}} \in (0,1)$ as a fixed point of the balance of information increment and losses from staleness. The discrepancy with the empirical ν_c^{emp} supports this decision — direct substitution of the simplest form of (11b) is rejected, which is recorded as the limit of its interpretive force. The derivation of a quantitative formula $\nu_c^{\mathrm{theor}}(\lambda, \sigma, K, \tau_{1/2}, \dot{M}_{\mathrm{refresh}})$ is an open problem § 8.5 main.

Active part of memory. In (11b) $|M_{\le t}|$ is the *active* part of memory, not the structural capacity limit $|M_{\le t}^{\mathrm{cap}}|$. In § 6.1 main: $|M_t| = k^2 = 64$ parameters of the estimator; this quantity grows with time in systems with expanding memory, and the numerical value of ν_c^{theor} depends on time. The empirical proxy ν_c^{emp} of § 6.1 main is ν_c^{op} by (7') main, not ν_c^{theor}; the working assumption of § 4.1 main on the stability of the regime threshold under a change of normalisation provides the transition between them with a possible numerical shift.

5.4 S5.4 The optimal τ^*_{reset} and its connection to dynamic regret

The optimal τ^*_{reset} maximises $\langle \eta_L \rangle$: frequent reset increases the payment, infrequent reset leads to nostalgic collapse (§ 5.3 main). The structural estimate:

$$\tau^*_{\mathrm{reset}} \sim \tau_E \cdot \log(|M_{\le t}^{\mathrm{cap}}| / \dot{M}_{\mathrm{refresh}} \tau_E),$$

where $|M_{\le t}^{\mathrm{cap}}|$ is the structural capacity limit of memory. The logarithmic dependence is a reflection of the standard form of the optimum for processes with exponential staleness and a linear refresh payment.

Biological analogy. The generation time of a taxon correlates with the rate of change of the econiche — an empirical regularity of population biology, consistent with (12) main up to a logarithmic factor. In mammals, in early embryogenesis and the germ line, an almost complete erasure of DNA methylation with re-establishment occurs (Reik 2007); this can be interpreted as a mechanism of τ^*_{reset} on the generational scale — a periodic reset of the nostalgic layer for the sake of restoring $\nu \to 0$ in a new generation.

Connection to dynamic regret bounds. In non-stationary stochastic optimisation (Besbes et al. 2015) (Theorem 2, stochastic setting) the regret-optimal length of the restart epoch has a *power-law* structure: the optimal length $\propto (T/V_T)^{2/3}$, where V_T is the variation of the target on the horizon T. This differs from the logarithmic form $\tau^*_{\mathrm{reset}} \sim \tau_E \log(\cdot)$ main §8.4(b): the main text gives a *logarithmic* form through a linear refresh payment with exponential staleness, whereas the BGZ regret-optimal restart is *power-law* $(T/V_T)^{2/3}$. Only the monotonicity in τ_E coincides (slower drift — rarer reset), not the functional form. In no-free-lunch (Wolpert 1996) a universal τ^*_{reset} is impossible, which is consistent with the conditionality of (12) main relative to the class of § 5.2 main.

5.5 S5.5 The analogy with spin glasses

(11b) main is a mean-field balance, structurally analogous to replica symmetry in the statistical mechanics of learning (Engel and Van den Broeck 2001). In the classical Hopfield model (Hopfield 1982) the critical load $\alpha_c \approx 0.138$ (Amit et al. 1985) is the threshold beyond which the associative memory loses the ability to retrieve the stored patterns without errors. Our ν_c^{theor} is the analogue in a non-equilibrium learning system: the threshold beyond which the nostalgic fraction exceeds a critical fraction and the predictive power $\eta_L(t)$ collapses.

The quantitative connection between ν_c^{theor} and the Hopfield α_c is a separate problem: α_c is the ratio of the number of stored patterns to the number of neurons in an equilibrium system with a fixed set of targets, ν_c^{theor} is the threshold ratio of the nostalgic fraction in a non-equilibrium system with a drifting latent parameter. The structural correspondence — both quantities reflect the limiting load on the memory capacity in systems with a fixed number of degrees of freedom; the dynamical characteristics, however, are substantially different.

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6 S6. Full technical details of § 6 main: PSP parameters, Robbins–Monro learner, wide-window control

This section assembles the technical details of the numerical illustrations of § 6.1 (PSP) and § 6.2 (OU) main, which in the main text are left in the form of a brief statement of results with a cross-ref here. Structure: S6.1 — full parameters of the PSP simulation (§ 6.1 main) and the MI estimator; S6.2 — simulation pseudocode and comparative analysis of estimators; S6.3 — the Robbins–Monro online learner of the OU simulation (§ 6.2 main) and its asymptotic Bayes-optimality; S6.4 — the numerical estimate of c for Robbins–Monro and the discrepancy with the prediction (8a); S6.5 — the wide-window negative control of the adiabatic scan and the discrepancy $C_v^{\text{static}}/C_v^{\text{predictive}}$; S6.6 — the empirical ν_c^{emp} and its status.

All reproducibility parameters (seed, library versions, directories) are S4 supplementary (simulation parameters for reproducibility); this section focuses on the methodological justification of the choice of parameters and the interpretation of the results.

6.1 S6.1 PSP simulation (§ 6.1 main): full parameters and estimator

Model. A finite-state Markov process with Poisson switches of the transition matrix. Alphabet $\Omega = \{1, \ldots, k\}$, $k = 8$. The matrix $P(t)$ is piecewise-constant on the intervals $[T_i, T_{i+1})$; the switch times $\{T_i\}$ form a Poisson stream with intensity λ. At time T_i the matrix is updated to a random stochastic $P^{(i)} \sim \text{Dirichlet}(\alpha = 0.3)$ by rows. $\alpha = 0.3$ gives $\langle \tau_{\text{relax}} \rangle \approx$ 2–3 steps and $\lambda \tau_{\text{relax}} \approx 3 \cdot 10^{-3} \ll 1$ with margin — slow-driving is satisfied.

Observer's memory. The observer system maintains a memory M_t consisting of an estimate of the current transition matrix $\hat{P}(t)$ on a sliding window of length τ_w with Laplace smoothing (pseudocount 1 per cell). The structural dimension of the memory $|M_t| = k^2 = 64$ parameters is the number of independent cells of the estimator; for the budget $N_{\max}(t) = N_0 + R \cdot t$ in the simulation the normalisation $k_B T \ln 2 = 1$ and $R = 1$ unit per step are adopted (the base operational payment plus Lemma 1 on the k^2 maintained parameters).

Parameters of the three regimes. Stationary regime: $\lambda = 0$ (the matrix does not change), $\tau_w = 100$, $T = 10^4$ steps, averaging over 80 realisations. Drift without reset: $\lambda = 10^{-3}$ (mean $\tau_E = 10^3$ steps between switches), $\tau_w \to \infty$ (the memory is not cleared, $\hat{P}(t)$ is averaged over the whole history), $T = 10^5$ steps, 60 realisations. Drift with reset: $\lambda = 10^{-3}$, $\tau_w \in \{100, 500, 2000\}$, $T = 10^5$, 60 realisations.

Grid of τ_w for Fig. 2. Eight points $\tau_w \in \{50, 100, 200, 500, 10^3, 2 \cdot 10^3, 5 \cdot 10^3, 10^4\}$ at $\lambda = 10^{-3}$, $T = 5 \cdot 10^4$, 40 realisations per point.

6.2 S6.2 Simulation pseudocode and comparative analysis of MI estimators

Pseudocode of the PSP simulation.

```
for each condition (λ, τ_w):

    initialise P_0, draw T_1 ~ Exp(λ)

    initialise memory: empirical_freq = matrix of zeros (k × k)

    for t = 0, 1, ..., T:

        if t == T_i: update P to a random P^(i) ~ Dirichlet(α=0.3, k rows),

                     draw T_{i+1}

        X_t = sample from P_t[X_{t-1}, :]

        update empirical_freq[X_{t-1}, X_t] += 1

        if t mod τ_w == 0 and τ_w < ∞: empirical_freq = 0  # reset

        P_hat(t) = empirical_freq normalised by rows (with Laplace smoothing)

        compute I_pred(t, τ) = MI(X_{t+1..t+τ}; P_hat(t))

        compute ν(t) = 1 - I_pred(t,τ) / I_pred^optimal(P_t, τ)

        accumulate N_max(t) = accumulated storage cost of empirical_freq + capacity growth
```

```
    η_L(t) = I_pred(t, τ) / N_max(t)

for each set (λ, τ_w): plot η_L(t), ν(t)
```

MI estimator: KL-detuned proxy. At prediction horizon τ the numerical estimate of the kernel definition (4) main is carried out through

$$\widehat{I}_{\text{pred}}(t,\tau) = I_{\text{opt}}(P(t),\tau) - \mathbb{E}_{X_t\sim\pi(t)}\big[D_{\text{KL}}\big(P(t)^\tau(\cdot \mid X_t) \,\|\, \hat{P}(t)^\tau(\cdot \mid X_t)\big)\big],$$

clipped at zero from below. Here $I_{\text{opt}}(P,\tau) = H(\pi) - \sum_i \pi_i H(P^\tau_{i,\cdot})$ is the predictive information of the optimal observer with the exact model; the KL penalty for the mismatch of $\hat{P}$ to the true P is subtracted. Connection to (4) main: for a Markov chain with a fixed estimator $\widehat{I}_{\text{pred}}(t,\tau)$ is majorised from above by the conditional MI $I(X_t; X_{t+\tau} \mid \hat{P}(t))$ (the KL penalty accounts for the maladaptation of the model as a lower bound on the loss of informativeness); the clipping at zero from below is ensured by the fact that the conditional MI is non-negative. On the ergodic plateau $\hat{P}(t) \to P(t)$ the KL penalty vanishes, and $\widehat{I}_{\text{pred}} \to I_{\text{opt}} = I(X_t; X_{t+\tau} \mid P_0)$, which is consistent with (4) main in the stationary limit.

Comparison with independent estimators. Alternative discrete MI estimates with bias correction (Treves and Panzeri 1995) and the continuous KSG estimator with k-NN, $k_{NN} = 3$ (Kraskov et al. 2004) give consistent results on the parameter grid of the PSP simulation. The consistency of three independent estimators (KL-detuned proxy, Treves–Panzeri plug-in, KSG) on the same trajectories confirms that the observed qualitative patterns (Fig. 1–3 main) are not an artefact of a specific MI estimation method.

Dome-shaped dependence $\langle\eta_L\rangle(\tau_w)$. The dome-shaped curve of Fig. 2 main with maximum $\langle\eta_L\rangle^* \approx 1.3\cdot 10^{-5}$ at $\tau_w^* \approx 200$ steps (plateau 200–500) is the non-stationary analogue of the bias-variance trade-off of classical learning theory: small τ_w — high variance of the estimator $\hat{P}$ (insufficient data), large τ_w — bias due to averaging over outdated regimes; the optimum τ_w^* is the balance, analogous to structural risk minimisation.

The position of the maximum $\tau_w^* \approx 0.2\,\tau_E$ is consistent with (12) main up to a logarithmic factor; the exact value of the optimum depends on fine parameters of the estimator (Laplace smoothing, the α of the Dirichlet).

6.3 S6.3 The Robbins–Monro online learner of the OU simulation (§ 6.2 main)

Maintenance of the online estimate. In the OU simulation (§ 6.2 main) the learner maintains an online estimate $\hat{\theta}(t)$ through the maximisation of the conditional log-likelihood with a decreasing step $\eta_t = c_0/\sqrt{t}$ ($c_0 = 3$): at each step, upon observing $(x_t \to x_{t+1})$, one performs

$$\hat{\theta}_{x_t,j}(t+1) = \hat{\theta}_{x_t,j}(t) + \eta_t \cdot \left[\mathbb{1}[x_{t+1} = j] - \hat{P}_{x_t,j}(t)\right]$$

for $j \neq x_t$. This is a diffusion-prior-Bayes approximation of the Bayesian update over the OU latent in the sense of § 5.2 main; the rate $1/\sqrt{t}$ corresponds to the concentration of the posterior along each of the K directions and is asymptotically Bayes-optimal for regular parametric models (Clarke and Barron 1990, § 3).

Polyak–Ruppert averaging note. The standard improvement of Robbins–Monro — Polyak–Ruppert averaging of the iterates — gives an asymptotically efficient estimate with smaller variance; in the present implementation it is not used, since the observed qualitative patterns (Fig. 4–5 main) are robust to the choice of Robbins–Monro variant up to log factors on the final variance. The use of Polyak–Ruppert would change the quantitative values of $\eta_L(t)$ by a constant factor, but not the qualitative picture of the collapse.

Structural equivalence to Bayes. Robbins–Monro with the indicated rate $\eta_t \propto 1/\sqrt{t}$ is structurally equivalent to an online approximation of the Bayesian update over the OU latent of § 5.2 main in the following sense: the post-cumulative distribution of the estimates $\hat{\theta}(t)$ as $t \to \infty$ concentrates around the conditional mean of the posterior with covariance of order $\mathcal{F}^{-1}/t$, where $\mathcal{F}$ is the Fisher information matrix of the parametric model (Clarke and Barron 1990). This is the justification of the choice of $\eta_t \propto 1/\sqrt{t}$ as functionally equivalent to the Bayesian learner.

6.4 S6.4 The numerical estimate of c for Robbins–Monro and the discrepancy with (8a)

Effective refresh rate. The adaptive step $\eta_t = c_0/\sqrt{t}$ with $c_0 = 3$ gives an effective refresh rate $\dot{M}^{\mathrm{eff}}_{\mathrm{refresh}}(t) \sim K \cdot \eta_t/\tau_E = Kc_0/(\tau_E\sqrt{t})$; for $K = 56$, $\tau_E = 10^3$, $t = 2 \cdot 10^4$ one obtains

$$c(t) = \frac{\dot{M}^{\mathrm{eff}}_{\mathrm{refresh}} \cdot \tau_E}{|M_{\leq t}|} \approx \frac{Kc_0}{|M_{\leq t}|\sqrt{t}} \approx \frac{56 \cdot 3}{56 \cdot \sqrt{2 \cdot 10^4}} \approx 0.02.$$

Discrepancy with the prediction of Lemma 2. Lemma 2 gives $\liminf \nu^{\mathrm{theor}}(t) \geq 1 - c \approx 0.98$. The observed $\nu(t > 10^4) = 0.82 \pm 0.03$ is *below* the predicted asymptote; the gap ≈ 0.16 is explained by (i) the finiteness of the observation time (the asymptotics (8a) main is the $\liminf$ as $t \to \infty$); (ii) the per-bit-uniformity assumption of Remark 5 (S5.1 supplementary), holding only approximately for the OU parametrisation of the logits with softmax nonlinearity (Remark 6). The full proof of convergence to $1 - c$ for the specific OU parametrisation is an open problem § 8.5 main.

6.5 S6.5 Wide-window negative control and the discrepancy $C_v^{\mathrm{static}}/C_v^{\mathrm{predictive}}$

Wide-window negative control. Upon extending the window of the joint fit of the adiabatic scan (§ S8.3) to $[10^3, T_{\mathrm{total}}]$ (instead of the fixed $[10^3, 10^4]$) the fit degenerates: for $\sigma^2/\lambda = 10^{-3}$ and $T_{\mathrm{total}} = 2 \cdot 10^5$ the coefficient $B/(K/2)$ grows to 2.0 — an artefact of the Robbins–Monro $1/\sqrt{t}$ bias of the learner on long horizons, not part of the

BNT effect. The window $[10^3, 10^4]$ is chosen for direct comparison between the points of the adiabatic series and is consistent with the transient condition $t \ll 1/\sigma^2$. This control records the sensitivity of the result to the choice of the fit window and prevents the interpretation of the monotone trend of $B/(K/2)$ as the result of selective presentation.

Numerical scan $B/(K/2) \in \{0.54, 0.78, 0.85\}$. Full results — § S8.3 (table of B, $B/(K/2)$, R^2, drift-share); S4.3 supplementary (scan parameters).

Discrepancy $C_v^{static}/C_v^{predictive}$. In the "drift without reset" regime (§ 6.1 main) at $\nu(t) \to 0.999$ one has $C_v^{\text{predictive}}(t) = (1-\nu(t)) \cdot C_v^{\text{static}}(t) \to 0$, whereas C_v^{static} remains bounded below by the size of the accumulated memory $\hat{P}(t)$. The ratio $C_v^{\text{static}}/C_v^{\text{predictive}} \to \infty$ is a numerical illustration of the diagnostic signal of nostalgia (§ 4.3 main): two proxies, agreeing in the stationary regime, diverge under observed drift. This operationalises Prediction 1 § 8.6 main: the crossing of ν_c^{op} is diagnosed through the divergence of the two proxies, without direct measurement of $\nu(t)$.

6.6 S6.6 The empirical ν_c^{emp} and its status

The empirical ν_c^{emp}, defined as the value of $\nu(t)$ at the moment when $\eta_L(t)$ falls to half of its peak in the drift-without-reset regime, equals $\nu_c^{\text{emp}} \approx 0.44$ for the base parametrisation of the PSP. This value is consistent with the structural statement of § 5.3 main on the existence of $\nu_c \in (0, 1)$ separating the adaptive regime and the collapse regime, but **does not claim a universal quantitative estimate**: ν_c^{emp} depends on the chosen operational definition (the half-life threshold of η_L) and on the parameters of the specific PSP surrogate.

Discrepancy with (11b). The direct balance substitution $\nu_c = \eta_M/(1+\eta_M) \approx 0.016$ in (11b) main is rejected empirically — a discrepancy of more than an order of magnitude. This supports the decision of § 5.3 main / S5.3 supplementary to treat (11b) as a structural balance condition, not as a quantitative predictor. The derivation of a universal formula $\nu_c(\lambda, \sigma, K, \tau_{1/2}, \dot{M}_{\text{refresh}})$ is an open problem § 8.5 main.

Stability under a change of normalisation. By DPI $\nu^{\text{op}}(t) \le \nu^{\text{theor}}(t)$ pointwise (§ 4.1 main; the smaller denominator $C_\mu^{\text{emp}} \le I_{\text{pred}}^{\text{opt}}$ in (7') gives a larger fragment and smaller nostalgia); consequently the empirical ν_c^{emp} is ν_c^{op}, a lower estimate for ν_c^{theor}. The working assumption of § 4.1 main on the monotonicity of the regime threshold under a change of normalisation provides the applicability of the empirical ν_c^{emp} as an operational surrogate of the theoretical ν_c^{theor} with a possible numerical shift, but without a qualitative displacement of the threshold.

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7 S7. Active inference: full formal correspondence between η_L and EFE, $\rho(t)$, caveats

This section assembles the formal details of § 7 main, moved here to lighten the main exposition. Structure: S7.1 — the heuristic correspondence $\Delta G^{\text{epist}} \leftrightarrow -\Delta I_{\text{pred}}$ (S7.1), the full justification with the conversion of units; S7.2 — the conjecture (S7.2) with its explicit hypothetical status and a discussion of the direction of formalisation; S7.3 — the categorical caveat $I_q(s';o') \neq I(X_t;X_{t+\tau})$ through the sufficiency of $\hat{P}(t)$; S7.4 — the context of sophisticated inference (Friston et al. 2021) and the Pearl/Friston blanket (Bruineberg et al. 2018; Bruineberg et al. 2022) demarcation; S7.5 — the rebuttal of Andrews (2021) / Williams (2020) through a cross-ref to paper #1; S7.6 — the informational rejuvenation metric $\rho(t)$ in detail with empirical interpretation.

7.1 S7.1 Heuristic correspondence: full justification

In the formulation of active inference (Friston et al. 2017a, eq. 3; Parr2022, ch. 7) the expected free energy of a policy π is reckoned in nats and is

$$G[\pi] = \mathbb{E}_{q(o',s'|\pi)}[\ln q(s' \mid \pi) - \ln p(o', s' \mid C)]\,,$$

where $p(o', s' \mid C)$ is the preference distribution with parameter C encoding the agent's goals. The canonical decomposition decomposes $G = -\mathbb{E}_q[D_{\mathrm{KL}}(q(s' \mid o', \pi)\|q(s' \mid \pi))] - \mathbb{E}_q[\ln p(o' \mid C)]$ into epistemic value (the information gain about hidden states) and pragmatic value (the proximity to preferences).

Conversion of units. G is measured in nats; I_{pred} in bits (Appendix A main). Conversion: $G_{[\mathrm{nats}]} = G_{[\mathrm{bits}]} \cdot \ln 2$; conversion into energy units through $\beta = 1/(k_B T)$ gives $E_G = k_B T \cdot G_{[\mathrm{nats}]}$. Similarly $E_{I_{\mathrm{pred}}} = k_B T \ln 2 \cdot (I_{\mathrm{pred}})_{[\mathrm{bits}]}$. For the increment over the policy window Δt:

$$k_B T \cdot \Delta G^{\mathrm{epist}}[\pi]\big|_{\Delta t} \approx -k_B T \ln 2 \cdot \Delta I_{\mathrm{pred}}\big|_{\Delta t}, \tag{S7.1}$$

up to an additive constant absorbing the pragmatic contribution and independent of π. Both sides have the dimension of energy (J); the minus sign reflects the convention "G is minimised under the maximisation of $\dot{I}_{\mathrm{pred}}$"; the equivalent dimensionless form is obtained by dividing by $k_B T$.

Hypothetical status. (S7.1) is fixed as a *heuristic correspondence*, not as a proven inequality. A proof of the strict inequality requires (i) careful handling of the time derivatives in a non-stationary generative model; (ii) a demonstration that the pragmatic contribution is indeed independent of π on the time scale Δt; (iii) accounting for the corrections from the categorical caveat S7.3. All three ingredients are absent in the existing literature on active inference in the non-stationary regime.

7.2 S7.2 Conjecture: hypothetical status

The hypothetical theorem is the direction in which (S7.1) can be turned into a strict inequality:

$$k_B T \cdot \Delta G^{\mathrm{epist}}[\pi]\big|_{\Delta t} \overset{\text{conj.}}{\geq} -k_B T \ln 2 \cdot \Delta I_{\mathrm{pred}}\big|_{\Delta t}, \tag{S7.2}$$

where the label "conj." marks the hypothetical status. The formal provability of (S7.2) is an open problem requiring careful handling of the time derivatives and a non-stationary generative model in the extended active inference framework (Parr et al. 2022).

Sophisticated inference as a direction of formalisation. Sophisticated inference (Friston et al. 2021) extends standard active inference to policies that account for post-observational updates of the future prior distributions; this is a natural framework for the careful handling of the time derivatives in a non-stationary generative model. A proof of (S7.2) in the sophisticated inference framework is a natural next step of the extension of the present work.

7.3 S7.3 The categorical caveat $I_q(s'; o') \neq I(X_t; X_{t+\tau})$

The epistemic value in (Friston2017) is the mutual information $I_q(s'; o' \mid \pi)$ between the hidden state s' and the observation o', whereas $I_{\text{pred}}(t, \tau)$ by (4) main is the MI between two consecutive observations X_t and $X_{t+\tau}$. These are categorically different objects: the former is the MI at a single point in time between the latent and the observation, the latter is the MI between two moments in time along the observed trajectory.

Bridge through the sufficiency of $\hat{P}(t)$. Under the interpretation $s' = \theta(t + \tau)$, $o' = X_{t+\tau}$ and the condition $X_{t+\tau} \perp \theta(t) \mid \hat{P}(t)$ (the statistical sufficiency of $\hat{P}(t)$ with respect to $X_{t+\tau}$), by the chain rule of MI one obtains

$$I_{\text{pred}}(t, \tau) \leq I_q(\theta(t + \tau); X_{t+\tau} \mid \hat{P}(t)).$$

The minimisation of the epistemic part of $G[\pi]$ under the budget $N_{\max}(t)$ *heuristically corresponds* to the local maximisation of $\eta_L(t)$ (in the sense of the heuristic correspondence § S7.1; the strict implication is not proven — the bridge gives an upper bound on I_{pred}, not an equality). The pragmatic component through the preference C is *outside* the thermodynamic scale; the full equivalence of η_L and G would require encoding the preference through the minimisation of $\dot{E}_{\text{actual}}$ — a nontrivial extension of the FEP.

The condition of correspondence is that $M_{\leq t}$ includes $q(s)$ and the history of observations (an extension of the generative model to the episodic level through hierarchical/motivated active inference (Pezzulo et al. 2018; Parr et al. 2022, ch. 7–8), where the higher levels of the hierarchy hold a slowly drifting context; parametric predictive coding (Bogacz 2017) does not introduce a separate episodic layer).

7.4 S7.4 Pearl/Friston blanket demarcation and the non-stationary regime

The connection between η_L and Friston's free energy principle (Friston 2010; Friston et al. 2017a) is formulated in (Andriishin 2026, § 4.4): the requirement of self-payment turns the FEP into a discriminative criterion, and η_L operationally closes the gap between the general formalism and the requirement that the energy belong to the modelling loop. The support is the distinction between the Pearl blanket (an epistemic instrument) and the Friston blanket (an ontological property with the requirement of self-payment) (Bruineberg et al. 2018; Bruineberg et al. 2022).

In the non-stationary regime the demarcation is preserved: the correspondence between η_L and EFE makes sense only for the Friston blanket; for systems with externalised payment (LLM-as-agent in the standard architecture) $\eta_L(t) \to 0$ — there is no correspondence. This is consistent with the structure of § 2.6 main: self-payment as a criterion of applicability of η_L is carried over to the non-stationary regime and to each of the three terms of (1) main.

Destructive argument of Bruineberg. The argument (Bruineberg et al. 2018; Bruineberg et al. 2022) against the ontologisation of the FEP without a self-payment criterion is a structural argument in favour of the present framework: η_L operationalises the requirement of self-payment and thereby answers the critique of Bruineberg in the sense developed in (Andriishin 2026, § 4.4a).

7.5 S7.5 Andrews (2021) / Williams (2020) cross-ref to paper #1

The rebuttal of the objections of Andrews (2021) (FEP as an empty tautology in the absence of independent criteria) and Williams (2020) (FEP as a correct but uninformative characterisation of any non-equilibrium system) is developed in (Andriishin 2026, § S4.4a); in the present work it is not repeated, but the structural correspondence is preserved: the non-stationary $\eta_L(t)$ closes the same gap as the stationary η_L — the requirement that the dissipative flow belong to the modelling loop through self-payment. The category of correspondence is *functional* (the common structure "optimisation of predictive information under a budget"), not ontological (the full identity of EFE and η_L as objects).

7.6 S7.6 The informational rejuvenation metric $\rho(t)$: in detail

The informational rejuvenation metric is the ratio of the refresh and growth rates of memory:

$$\rho(t) = \frac{\dot{M}_{\text{refresh}}(t)}{\dot{M}_{\text{grow}}(t)}.$$

Interpretation of the poles. $\rho \gg 1$ — refresh outpaces capacity growth: the refresh fraction dominates, the nostalgia ν remains low (the adaptive regime § 5.3 main). $\rho \ll 1$ — capacity growth outpaces refresh: the archival layer $M_{<t}$ accumulates faster than the current layer manages to be refreshed, and the nostalgia grows monotonically (the nostalgic-collapse regime).

Epistemic status of ρ. (13) main is a diagnostic independent of the direct measurement of $\nu(t)$: $\dot{M}_{\text{refresh}}$ and $\dot{M}_{\text{grow}}$ are operationally measurable in systems where the direct estimate of $\nu(t)$ through (7) main is unavailable. In a bacterium, $\dot{M}_{\text{refresh}}$ is the frequency of CheB-mediated demethylation of receptors; $\dot{M}_{\text{grow}}$ is the frequency of replication with the addition of new copies of receptors. In an LLM-corp, $\dot{M}_{\text{refresh}}$ is the frequency of continual pre-training or RAG update of the index; $\dot{M}_{\text{grow}}$ is the frequency of expansion of the corporate corpus.

Empirical calibration. The numerical thresholds ρ_c are a separate study. To a first approximation $\rho \sim 1$ corresponds to the balance condition (11b) main; the exact relation $\rho_c(\lambda, \sigma, K, \tau_{1/2})$ depends on the same parametric structure as ν_c^{theor} (S5.3 supplementary), and inherits its open status.

Epistemic-pragmatic decomposition (optional). In terms of active inference $\rho(t)$ can be interpreted as the balance of epistemic impulse (the increase of epistemic value through $\dot{M}_{\text{refresh}}$) and pragmatic investment (the expansion of the policy horizon through $\dot{M}_{\text{grow}}$). This decomposition is a natural consequence of the correspondence S7.1; its formal development requires the sophisticated inference framework S7.2.

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8 S8. Adiabatic asymptotics of L_{excess} as an open conjecture

This section contains the full development of the open conjecture on the adiabatic asymptotics of the cumulative excess-loss, to which main § 8.1, § 8.5 refer as the main open quantitative problem of the work. Structure: S8.1 — the statement of the conjecture and its theoretical motivation; S8.2 — the conditions of applicability and the separation of the two adiabatic regimes; S8.3 — the parametric scan and the numerical support.

The conversion of units for L_{excess} (nats $\leftrightarrow$ bits) is Appendix A main: L_{excess} is measured in **nats** (as in the original BNT literature (Bialek et al. 2001, § VI) and the MDL formulation); the explicit conversion $L^{[\text{bits}]}_{\text{excess}}(t) = L^{[\text{nats}]}_{\text{excess}}(t)/\ln 2$ is applied when comparing with the main text.

8.1 S8.1 Statement of the conjecture and theoretical motivation

An alternative theoretical characterisation of the accumulation of information by a learning system about the latent parameter of the environment is the cumulative excess-loss of the learner relative to the oracle:

$$L_{\text{excess}}(t) := \sum_{s\le t}\big[\ln p\big(X_s \mid X_{s-1},\, \theta(s)\big) - \ln \hat{p}\big(X_s \mid X_{s-1},\, \hat{\theta}(s)\big)\big]. \tag{S8.1}$$

Here $\theta(s)$ is the true value of the latent parameter of the environment at step s, $\hat{\theta}(s)$ is the learner's estimate from the history of observations; $L_{\text{excess}}(t)$ is measured in nats. The operational status of (S8.1): this quantity is directly computable in simulations with a known ground truth $\theta(s)$, but in real systems (biosphere, LLM-corp, bacterium) the latent parameter $\theta(s)$ is in principle inaccessible — L_{excess} remains counterfactual, theoretically related to the operationally measurable one-step $I_{\text{pred}}(t,\tau)$ (4) main, but not equivalent to it.

Connection to (4) main and the thermodynamics of memory. $L_{\text{excess}}(t)$ is the cumulative Bayesian regret of the learner relative to the oracle in the sense of (Rissanen 1986; Clarke and Barron 1990); for regular parametric models it is asymptotically related to $I(M_{\le t};\theta(t))$ through the theorem on the sufficient statistic in the sense of (Cover and Thomas 2006, ch. 2.9). The difference: L_{excess} is the *integral* information about the latent accumulated by time t; (4) main is the *density* of the information flow through the current step, bounded by $\ln k$.

Conjecture S8.1 (adiabatic asymptotics). *In the adiabatic OU limit* $\sigma^2/\lambda \ll 1$ *under ideal Bayesian learning the cumulative excess-loss* $L_{excess}(t)$ *(S8.1) grows as* $(K/2)\ln(\lambda t)$ *— the Class II asymptotics (Bialek et al. 2001, § VI):*

$$L_{\text{excess}}^{\text{Bialek}}(t) \sim \frac{K}{2}\ln(\lambda t), \tag{S8.2}$$

where $K = k(k-1)$ *is the dimension of the logits. If this secondary metric is adopted as the non-stationary analogue of predictive information and one defines through it* $\eta_L^{excess}(t) := L_{excess}(t)/N_{max}(t)$*, then in the leading order as* $t \to \infty$ *in the adiabatic OU limit* $\sigma^2/\lambda \ll 1$ *under an ideal Bayesian learner and a linear growth of the denominator* $N_{max}(t) \sim Rt$*:*

$$\dot{\eta}_L^{\text{excess}}(t) \sim -\frac{K\ln(\lambda t)}{2Rt^2}. \tag{S8.3}$$

Status of the conjecture. (S8.2) is the known Class II BNT asymptotics (Bialek et al. 2001) for regular parametric models; the coefficient $K/2$ arises in several paradigms of statistical learning theory for regular parametric models of dimension K: MDL (Rissanen 1986; Gr"unwald 2007) and BIC (Schwarz 1978) through the Laplace approximation on the posterior; PAC-Bayes (McAllester 1999; Catoni 2007) for a Gaussian prior; online Newton-step regret (Cesa-Bianchi and Lugosi 2006; Hazan et al. 2007) for exp-concave loss (the formal class to which the logarithmic bound (Hazan et al. 2007) applies; self-concordant is a narrower property, not used in Hazan et al. 2007). (S8.3) is its direct algebraic consequence under the assumption of a linear growth of the denominator. The hypothetical status pertains to two aspects: (i) the convergence of the coefficient in the fit of L_{excess} to $K/2$ in realisable parametrisations of OU at finite σ; (ii) the applicability of the result to the operationally measurable efficiency $\eta_L(t)$ (10) main, rather than to the secondary η_L^{excess}.

8.2 S8.2 Conditions of applicability and the separation of the two adiabatic regimes

Conjecture S8.1 requires the *simultaneous* satisfaction of two conditions (see § 5.2 main):

1. **Slow-driving** $\lambda\tau_{\text{relax}} \ll 1$ — necessary for the applicability of Lemma 2 § 4.4 main and the quasi-stationary interpretation of $\theta(t)$.
2. **OU-concentration** $\sigma^2/\lambda \ll 1$ — necessary precisely for (S8.2), not for (4) main: the concentration of the posterior over $\theta(t)$ must outpace the proper drift of $\theta(t)$.

Additionally: (iii) boundedly growing memory $|M_{\leq t}| = o(t/\ln t)$ (for the linearity of the denominator $N_{\max} \sim Rt$); (iv) the OU parametrisation of precisely the logits (the softmax form ensures the regularity of the parametric model); (v) an ideal Bayesian learner (the refresh payment is in $\dot{E}_{\text{actual}}^{\text{curr}}$, not in the nostalgic layer).

8.3 S8.3 Parametric scan and numerical support of the conjecture

The numerical study of (S8.2) is a parametric scan of the OU simulation over σ^2/λ with direct measurement of (S8.1). The implementation is `simulations/markov_drift_ou_iinf_adiab/`.

Model. The OU parametrisation of § 5.2 / § 6.2 main unchanged ($k = 8$, $\lambda = 10^{-3}$, $K = 56$, Euler–Maruyama $\Delta t = 1$); the ideal Bayesian learner is approximated by Robbins–Monro $\eta_t = c/\sqrt{t}$, $c = 3$. $L_{\text{excess}}(t)$ is measured by (S8.1) directly — it is not bounded by $\ln k$, and grows asymptotically as (S8.2).

Parametric scan. Three values of σ at fixed $\lambda = 10^{-3}$: $\sigma \in \{0.01, 0.003, 0.001\}$, giving $\sigma^2/\lambda \in \{0.1, 9 \cdot 10^{-3}, 10^{-3}\}$ respectively. Transient window $1/\sigma^2 \in \{10^4, 1.1 \cdot 10^5, 10^6\}$; simulation time $T \in \{5 \cdot 10^4, 5 \cdot 10^4, 2 \cdot 10^5\}$ within the transient window for all cases. Each point is the average over 25 independent realisations of the OU process; seed `SEED = 20260524`. Joint fit $L_{\text{excess}}(t) = A \cdot t + B \cdot \ln(\lambda t) + C$ on the window $t \in [10^3, 10^4]$ (fixed for direct comparison across σ).

Results. *Fig. 6: B vs σ^2/λ*

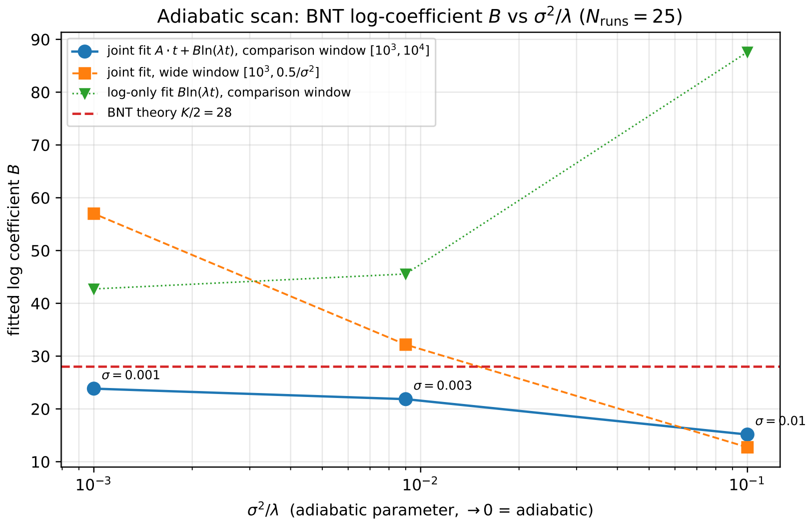
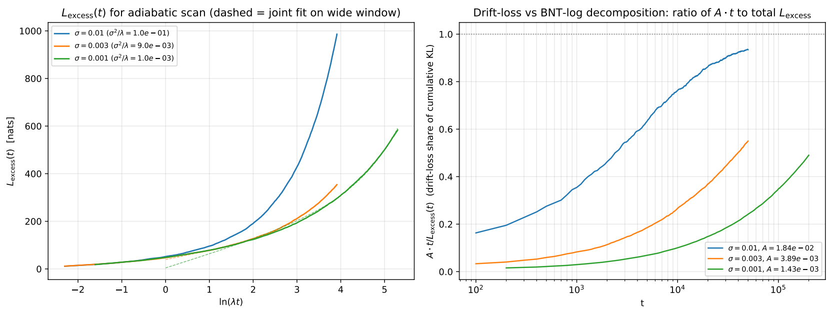


Fig. 7: family of $L_{excess}(t)$ for three σ



σ	σ^2/λ	B	$B/(K/2)$	R^2	drift-share at $t = 10^4$
0.01	10^{-1}	15.15	0.54	0.9998	76%
0.003	$9 \cdot 10^{-3}$	21.85	0.78	0.9996	27%
0.001	10^{-3}	23.84	0.85	0.9995	10%

The coefficient B grows monotonically as $\sigma^2/\lambda \to 0$, giving $B/(K/2) = 0.54, 0.78, 0.85$ respectively; $R^2 \geq 0.9995$ at all points, the drift-share $A \cdot t/L_{\text{excess}}(t)$ falls from 76% to 10%.

Wide-window negative control. Upon extending the window of the joint fit to $[10^3, T_{\text{total}}]$ (instead of the fixed $[10^3, 10^4]$) the fit degenerates: for $\sigma^2/\lambda = 10^{-3}$ and $T_{\text{total}} = 2 \cdot 10^5$ the coefficient $B/(K/2)$ grows to 2.0 — an artefact of the Robbins–Monro $1/\sqrt{t}$ bias of the learner on long horizons, not part of the BNT effect. The window $[10^3, 10^4]$ is chosen for direct comparison between the points of the adiabatic series and is consistent with the transient condition $t \ll 1/\sigma^2$. This control records the sensitivity of the result to the choice of the fit window and prevents the interpretation of the monotone trend of $B/(K/2)$ as the result of selective presentation.

Interpretation. The functional form $L_{\text{excess}}(t) = A \cdot t + B \ln(\lambda t) + C$ is extracted with $R^2 \geq 0.9995$; the coefficient B grows monotonically from 0.54 to 0.85 $\cdot K/2$ as σ^2/λ goes from 10^{-1} to 10^{-3}. **This is not a proof of convergence $B \to K/2$**; the asymptotic limit B_∞ remains an open question. The pre-registered protocol for confirming the conjecture (S8.2) requires a minimum of 5–6 scan points with an explicit analytical form of the finite-σ correction $B(\epsilon) = (K/2)(1 - g(\epsilon))$.

The relegation of (S8.2) and (S8.3) to the status of an open conjecture reflects the current state of the theoretical justification, not a final verdict: the numerical support of the functional form and the monotone convergence to $K/2$ are a nontrivial empirical regularity, deserving further analytical investigation (see § 8.5 main).

Reproducibility. The full implementation is `simulations/markov_drift_ou_iinf_adiab/` (`model.py`, `main.py`, `README.md`); the figures are in `paper/figs/Fig6.pdf` and `paper/figs/Fig7.pdf`; the summary log is `simulations/markov_drift_ou_iinf_adiab/results_summary.{txt,json}` and `run.log`. Seed `SEED = 20260524`; running `python main.py` (~1 minute on a modern laptop) reproduces all numerical values.

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9 S9. Details of §§ 2.1 and 2.3 main: three-faces comparison and majority-vote derivation

This section assembles the dense statistical-mechanics apparatus moved out of §§ 2.1, 2.3 main to lighten the main exposition. S9.1 — the full comparison of decomposition (1) main with the three-faces / Hatano–Sasa classification of entropy production; S9.2 — the full derivation of the majority-vote variant of Lemma 1 (equation (2') main).

9.1 S9.1 Comparison of decomposition (1) main with three-faces / Hatano–Sasa

Decomposition (1) main is *thematically parallel* to the three-faces decomposition of entropy production (Esposito and Van den Broeck 2010) and the Hatano–Sasa decomposition (Hatano and Sasa 2001), but does not reduce to it automatically. In the canonical treatment the housekeeping component maintains a fixed NESS under stationary control parameters, the excess component — the change of the NESS under a slow change of parameters. The correspondence of the components of (1) main:

- $\dot{E}_{\text{store}}$ for maintaining a specific information state against relaxation to symmetric equilibrium belongs to the *excess* class (relaxation-driven), not to housekeeping: it is not a fixed NESS that is maintained, but a specific non-equilibrium information state relaxing to symmetric equilibrium.
- $\dot{E}^{\text{curr}}_{\text{actual}}$ upon re-tuning $\hat{P}(t)$ to the drift of the environment also belongs to the excess class (the change of the target state under slow drift of the parameters).
- the refresh operations of Lemma 1 — a series of non-equilibrium protocols on top of this, reducible neither to purely housekeeping, nor to purely excess.
- $\dot{E}_{\text{grow}}$ — a structural rearrangement (constraint-driven), going beyond the canonical three-part decomposition: the addition of new degrees of freedom changes the state space itself, and not only the distribution on it.

The exact embedding of the non-stationary information decomposition into the three-faces apparatus is an open problem (§ 8.5 main).

9.2 S9.2 Full derivation of the majority-vote variant of Lemma 1 (equation (2') main)

In the limit $\varepsilon \to 0$ the polynomial bound (2) main $\sim 1/\varepsilon$ is replaced by the majority-vote variant through redundant coding. For n copies of a bit with independent errors $\varepsilon < 1/2$ the majority vote errs only when no fewer than $n/2$ copies are flipped; the probability of this tail of the binomial $\mathrm{Bin}(n, \varepsilon)$ is bounded by the Chernoff bound

$$\varepsilon^{\text{eff}} \;=\; \Pr[\mathrm{Bin}(n,\varepsilon) \geq n/2] \;\leq\; \exp\!\big(-n\, D(\tfrac{1}{2} \,\|\, \varepsilon)\big), \qquad D(\tfrac{1}{2} \,\|\, \varepsilon) = \tfrac{1}{2} \ln \frac{1}{4\varepsilon(1-\varepsilon)},$$

where $D(\frac{1}{2}\,\|\,\varepsilon)$ is the KL divergence of the Bernoulli Ber(1/2) from Ber(ε). Here ε in $D(\frac{1}{2}\,\|\,\varepsilon)$ is the *raw per-copy error* (the probability of a flip on one copy of a bit, a physical parameter of the carrier), whereas ε^{eff} is the *target* (achievable) effective error of the decoded bit after the majority vote; it is precisely the difference of these two quantities that ensures the exponential suppression $\varepsilon^{\text{eff}} \le e^{-nD(\frac{1}{2}\|\varepsilon)}$ at fixed $\varepsilon < 1/2$. Hence, to achieve a target effective error ε^{eff} it suffices to have

$$n \sim \frac{\ln(1/\varepsilon^{\text{eff}})}{D(\frac{1}{2}\,\|\,\varepsilon)}$$

copies — a *logarithmic* number of copies in $1/\varepsilon^{\text{eff}}$ (rather than linear in $1/\varepsilon$, as in naive refresh). Each of the n copies degrades independently with the same $\tau_{1/2}$, so the refresh rate for the majority vote preserves the factor $t/\tau_{1/2}$ (the number of refresh cycles), but the payment per cycle is logarithmic in $1/\varepsilon$, not polynomial:

$$E^{(1),\text{maj}}_{\text{store}}(t) \;\ge\; C \cdot \frac{t}{\tau_{1/2}} \cdot k_B T \ln(1/\varepsilon), \qquad C = O(1), \tag{2'}$$

with a constant C depending on the majority-vote scheme (Sagawa–Ueda extension Sagawa and Ueda 2009; Parrondo et al. 2015). Structurally, (2) main and (2') coincide in their linear growth with $t/\tau_{1/2}$ — the number of refresh cycles; the difference is in the dependence on ε: polynomial in naive vs logarithmic in majority-vote. The class of paradigm-case systems to which the original (2) main applies: ferromagnetic domains of warm media and systems with naive refresh; for biological systems with feedback repair (2') is used, with the logarithmic dependence on ε. Feedback-assisted refresh (DNA mismatch repair) additionally mitigates the bound by Sagawa–Ueda $\langle W\rangle \ge k_B T(\Delta S - I_{\text{meas}})$ (Sagawa and Ueda 2009): a *structured* bit with conditional entropy $H(\varepsilon) = -\varepsilon\ln\varepsilon - (1-\varepsilon)\ln(1-\varepsilon) \ll \ln 2$ at $\varepsilon \ll 1$ is erased, and in the limit of ideal measurement the cost of refresh goes to zero — the asymptotics of autonomous Maxwell demons (Koski et al. 2014; Mandal and Jarzynski 2012; Barato and Seifert 2014; Bauer et al. 2014).

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